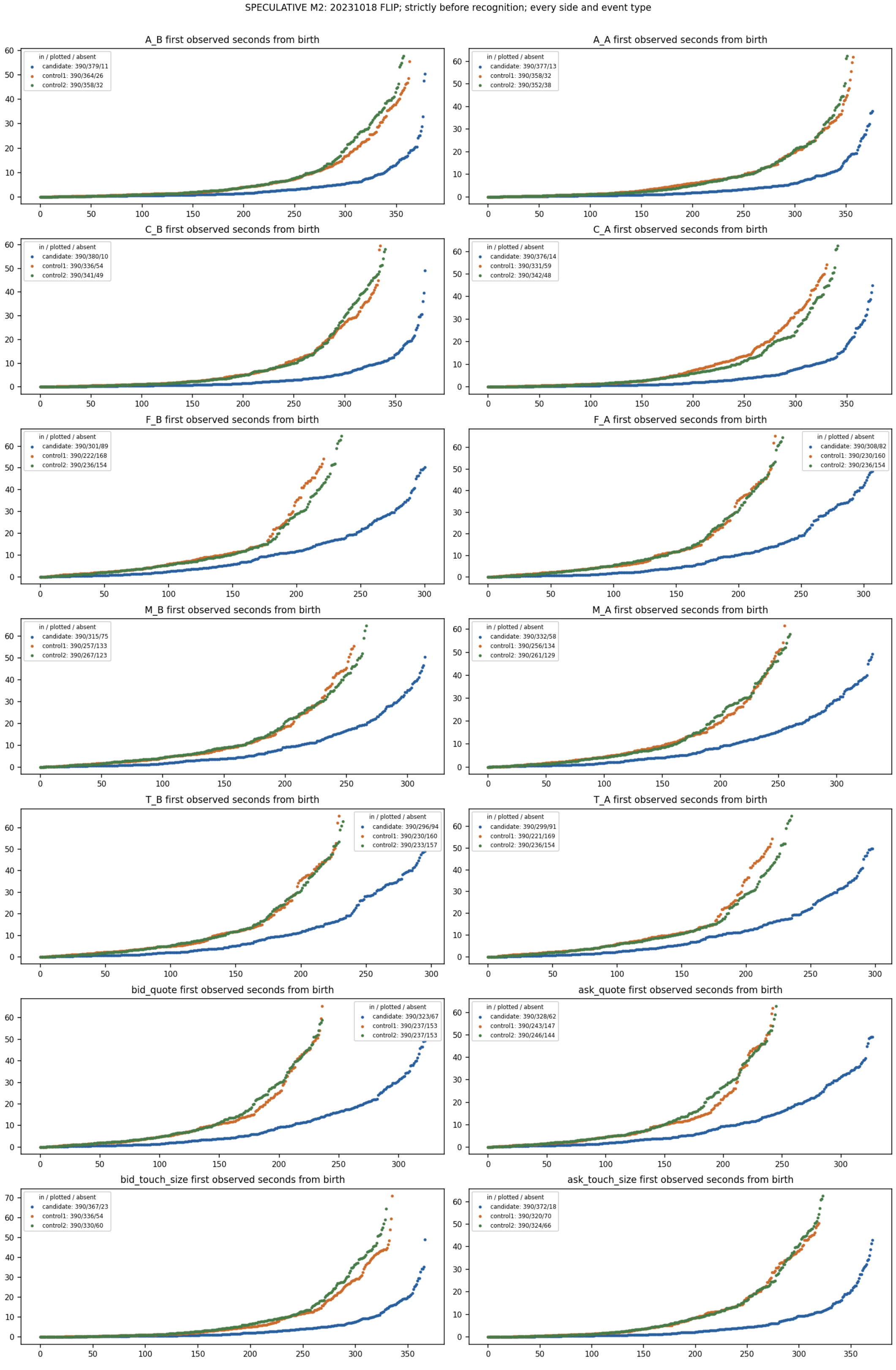
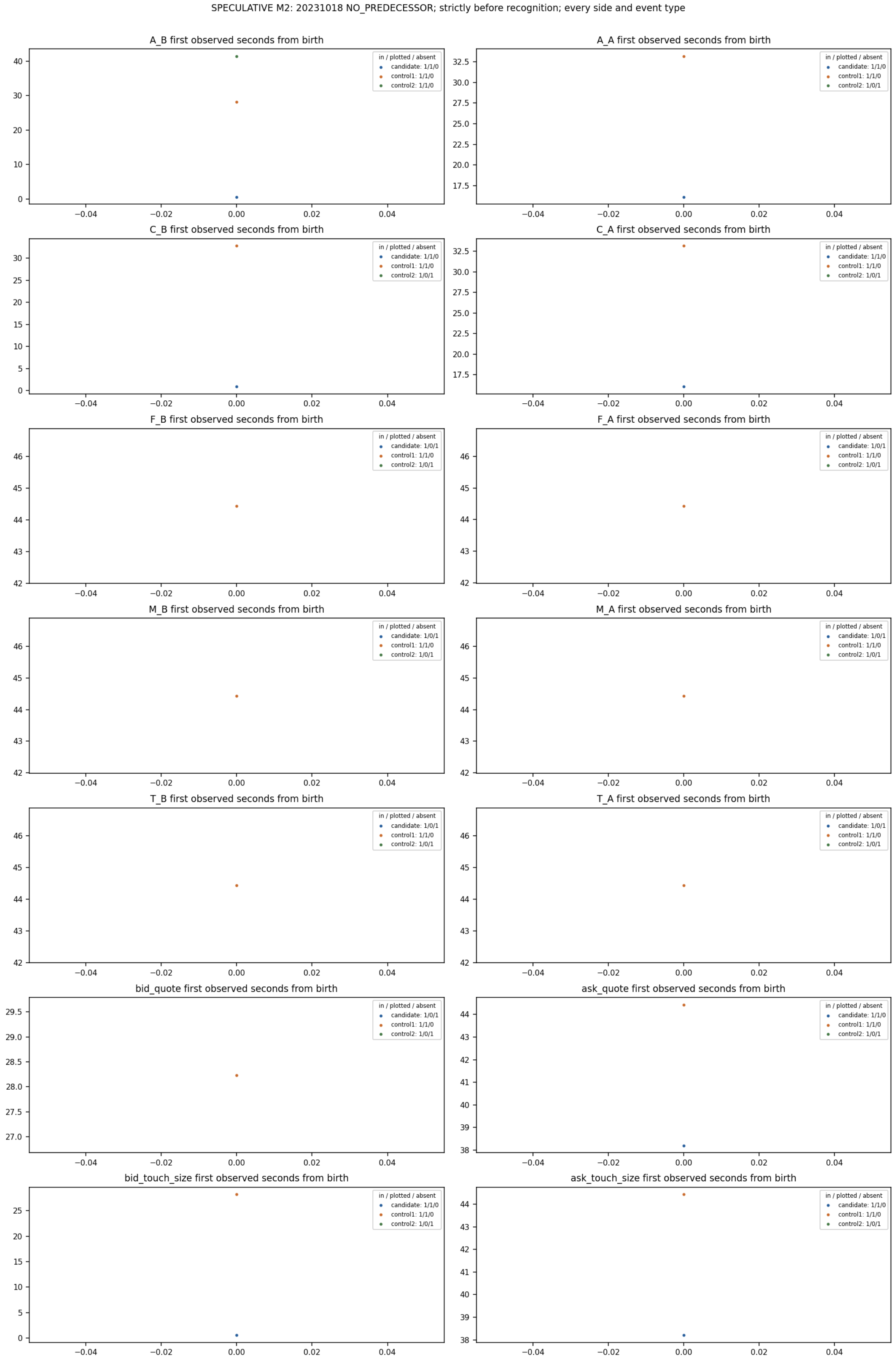
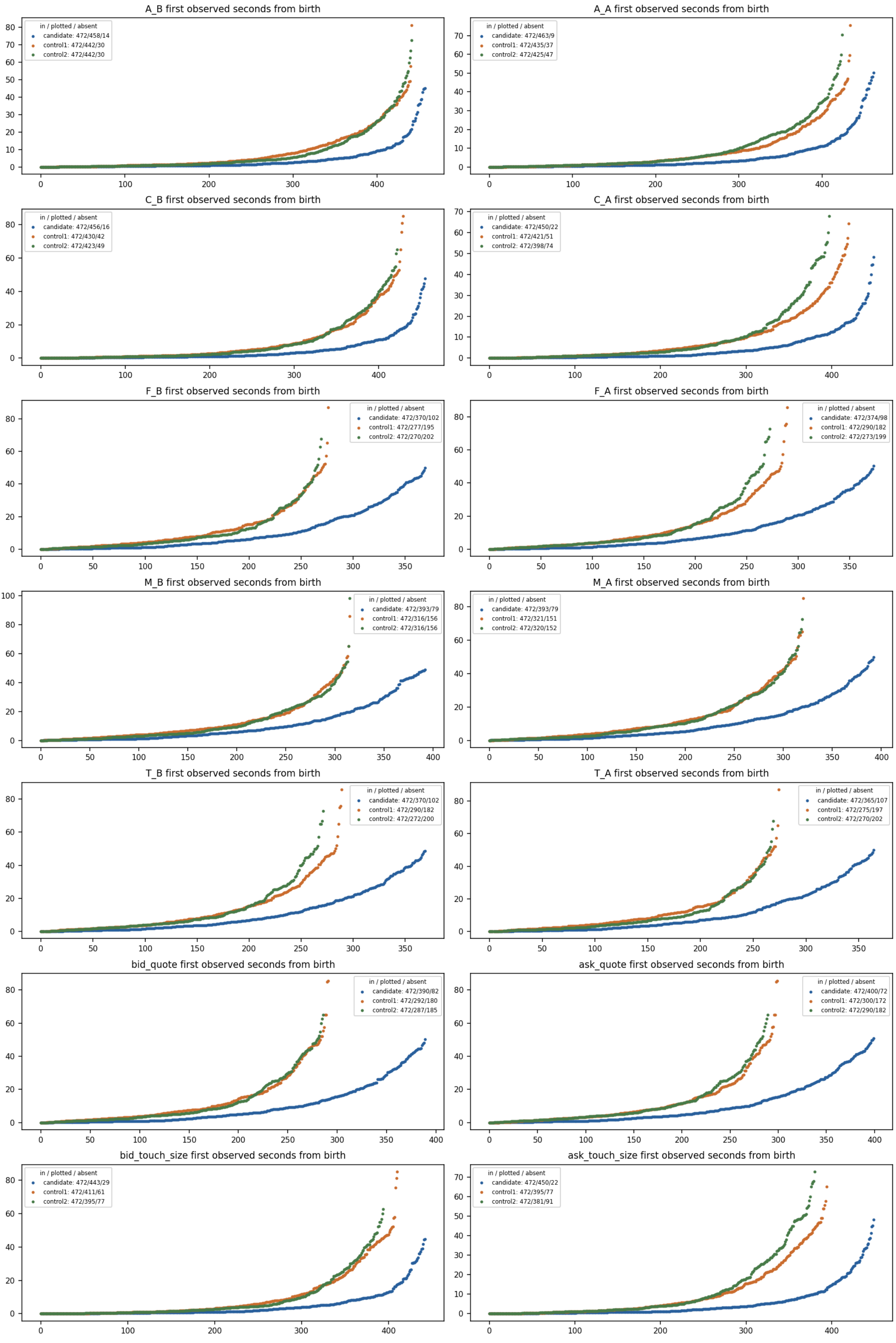
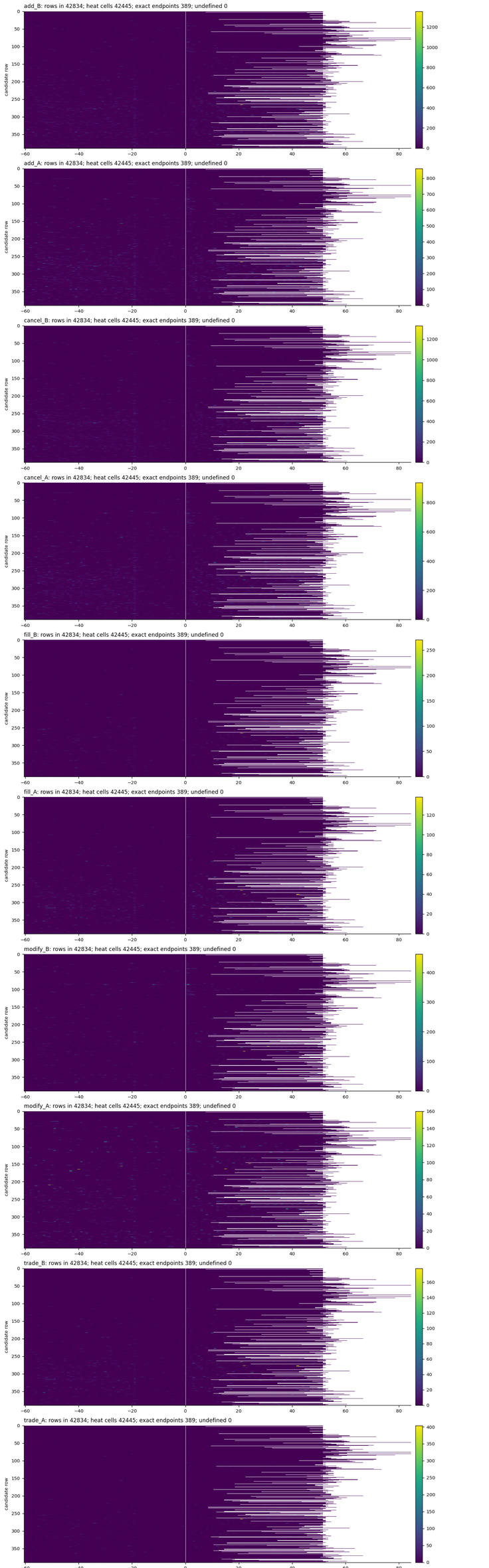

SPECULATIVE M2: 20231018 FLIP; strictly before recognition; every side and event type
A_B first observed seconds from birth
in / plotted / absent
candidate: 390/379/11
control1: 390/364/26
control2: 390/358/32
A_A first observed seconds from birth
in / plotted / absent
candidate: 390/377/13
control1: 390/358/32
control2: 390/352/38
C_B first observed seconds from birth
in / plotted / absent
candidate: 390/380/10
control1: 390/336/54
control2: 390/341/49
C_A first observed seconds from birth
in / plotted / absent
candidate: 390/376/14
control1: 390/331/59
control2: 390/342/48
F_B first observed seconds from birth
in / plotted / absent
candidate: 390/301/89
control1: 390/222/168
control2: 390/236/154
F_A first observed seconds from birth
in / plotted / absent
candidate: 390/308/82
control1: 390/230/160
control2: 390/236/154
M_B first observed seconds from birth
in / plotted / absent
candidate: 390/315/75
control1: 390/257/133
control2: 390/267/123
M_A first observed seconds from birth
in / plotted / absent
candidate: 390/332/58
control1: 390/256/134
control2: 390/261/129
T_B first observed seconds from birth
in / plotted / absent
candidate: 390/296/94
control1: 390/230/160
control2: 390/233/157
T_A first observed seconds from birth
in / plotted / absent
candidate: 390/299/91
control1: 390/221/169
control2: 390/236/154
bid_quote first observed seconds from birth
in / plotted / absent
candidate: 390/323/67
control1: 390/237/153
control2: 390/237/153
ask_quote first observed seconds from birth
in / plotted / absent
candidate: 390/328/62
control1: 390/243/147
control2: 390/246/144
bid_touch_size first observed seconds from birth
in / plotted / absent
candidate: 390/367/23
control1: 390/336/54
control2: 390/330/60
ask_touch_size first observed seconds from birth
in / plotted / absent
candidate: 390/372/18
control1: 390/320/70
control2: 390/324/66


SPECULATIVE M2: 20231018 NO_PREDECESSOR; strictly before recognition; every side and event type
A_B first observed seconds from birth
A_A first observed seconds from birth
C_B first observed seconds from birth
C_A first observed seconds from birth
F_B first observed seconds from birth
F_A first observed seconds from birth
M_B first observed seconds from birth
M_A first observed seconds from birth
T_B first observed seconds from birth
T_A first observed seconds from birth
bid_quote first observed seconds from birth
ask_quote first observed seconds from birth
bid_touch_size first observed seconds from birth
ask_touch_size first observed seconds from birth
in / plotted / absent
candidate: 1/1/0
control1: 1/1/0
control2: 1/0/1
candidate: 1/0/1
control2: 1/0/1

SPECULATIVE M2: 20231018 SAME; strictly before recognition; every side and event type

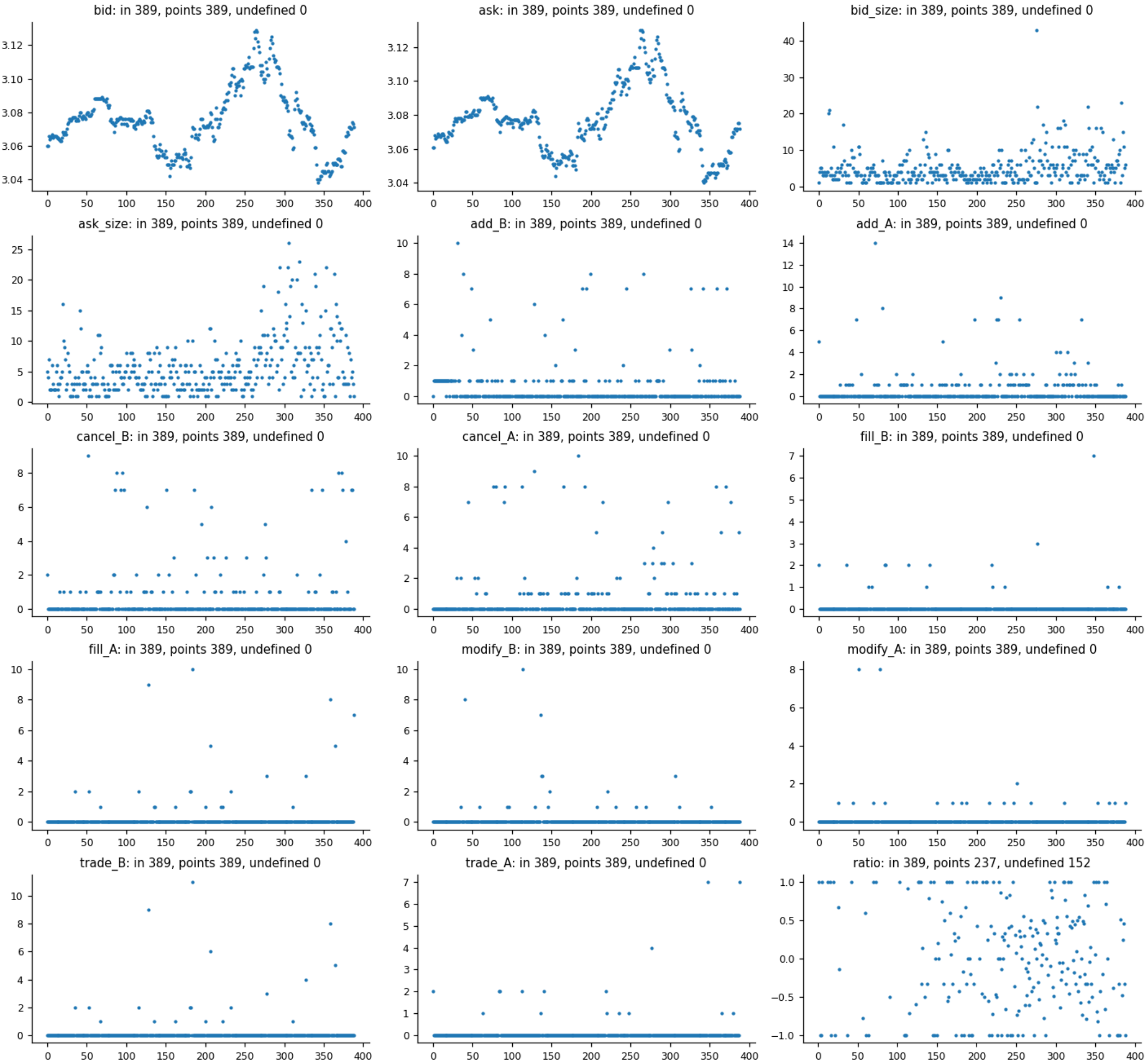


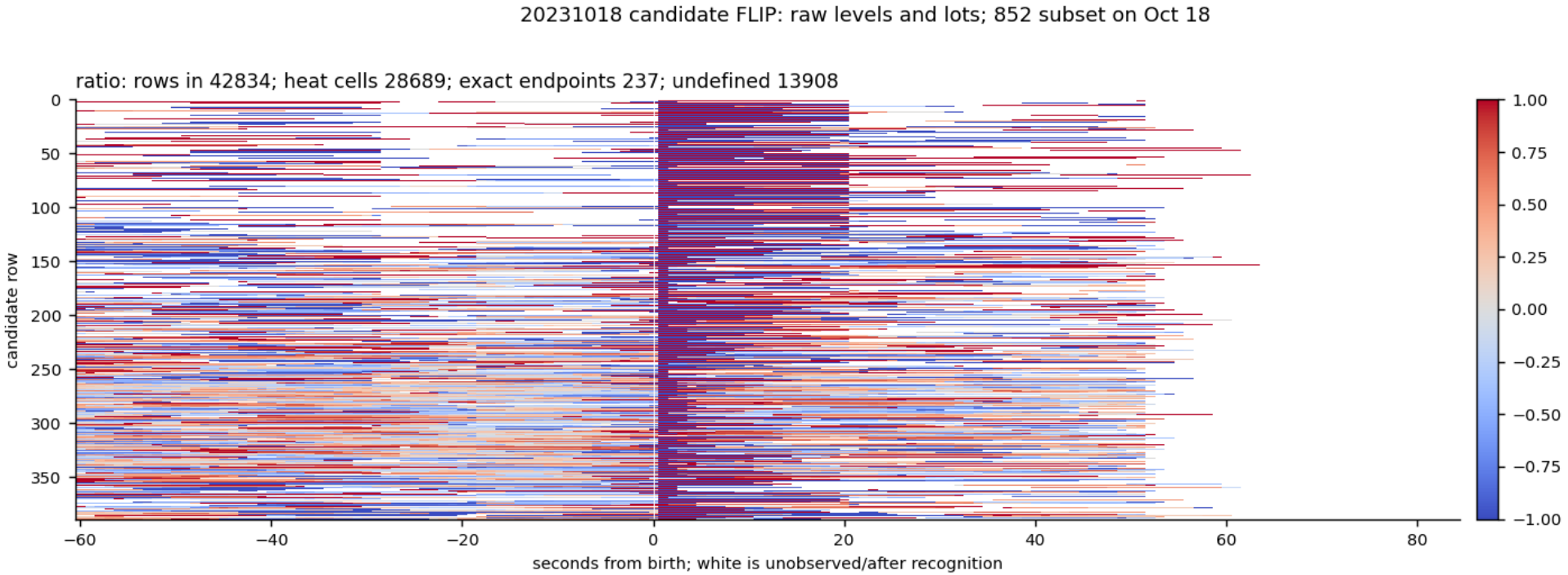
20231018 candidate FLIP: raw levels and lots; 852 subset on Oct 18



20231018-candidate-FLIP-endpoints.png

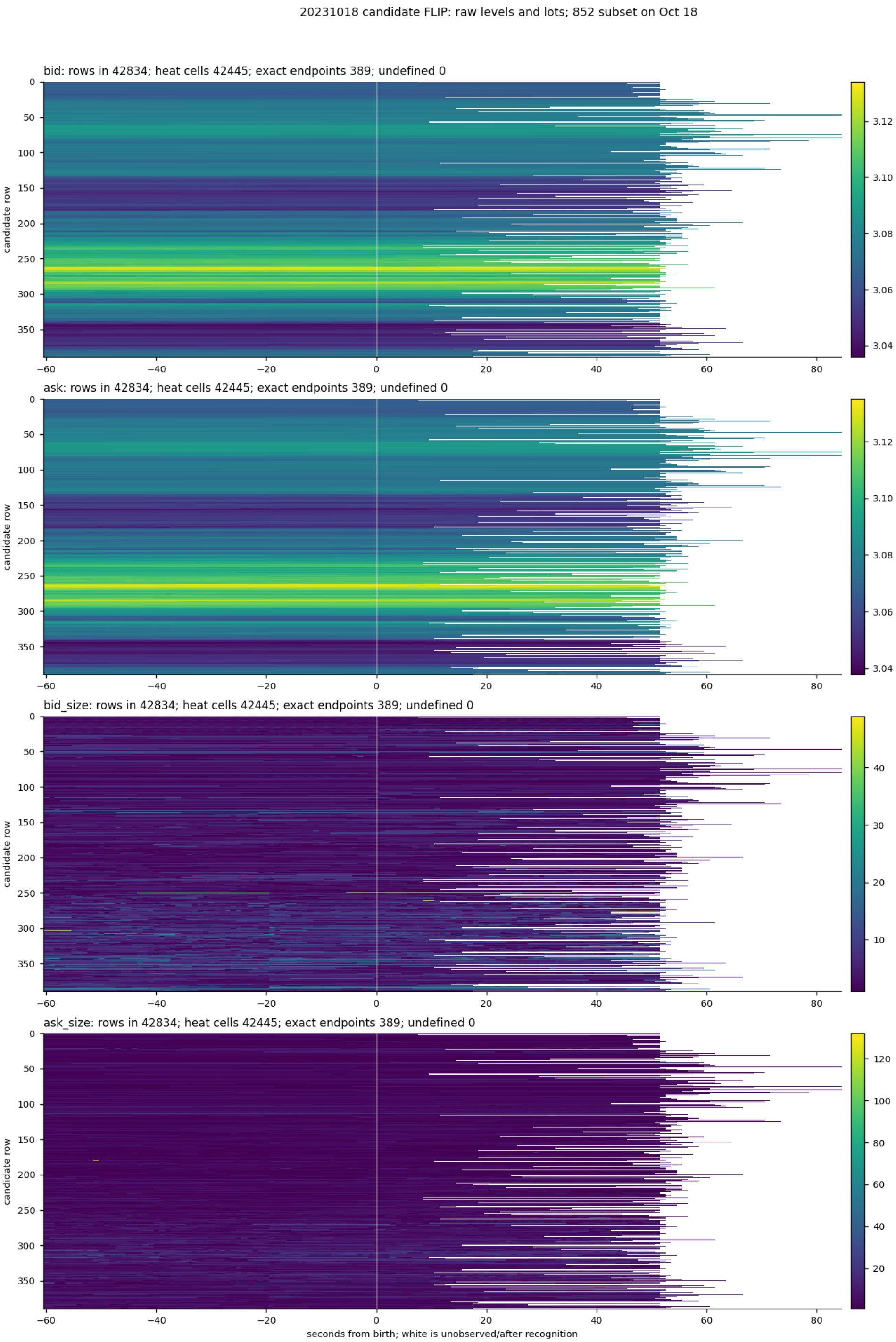
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.




20231018 candidate FLIP: raw levels and lots; 852 subset on Oct 18
ratio: rows in 42834; heat cells 28689; exact endpoints 237; undefined 13908
candidate row
0
50
100
150
200
250
300
350
−60
−40
−20
0
20
40
60
80
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

20231018-candidate-FLIP-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

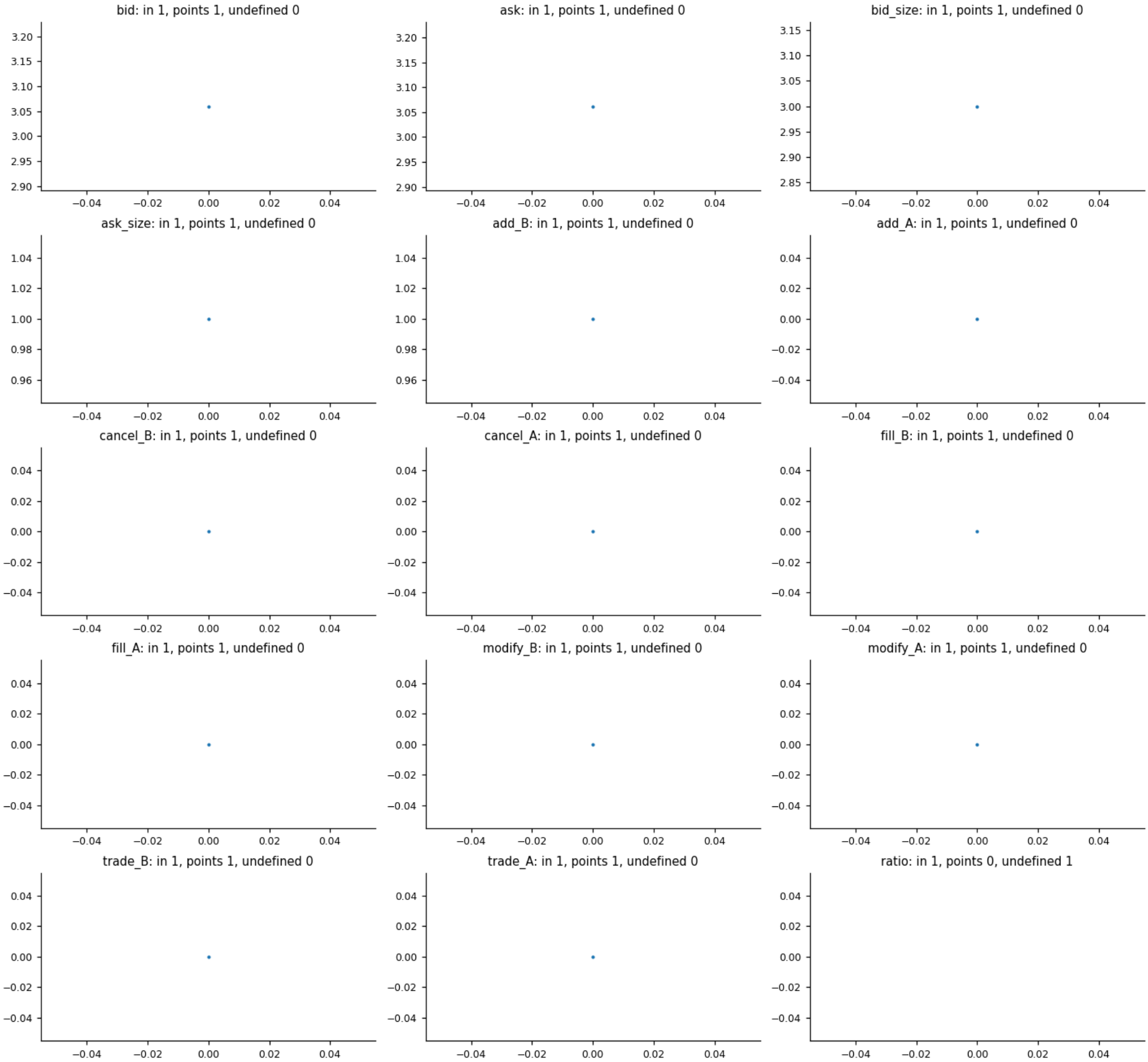


20231018-candidate-NO_PREDECESSOR-actions.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

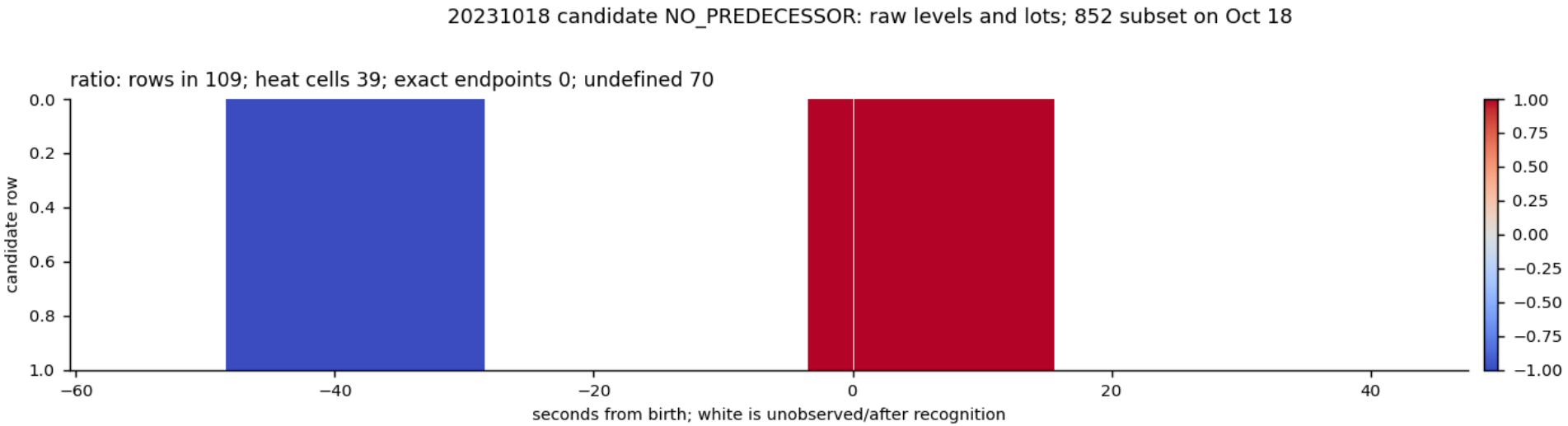
20231018-candidate-NO_PREDECESSOR-endpoints.png

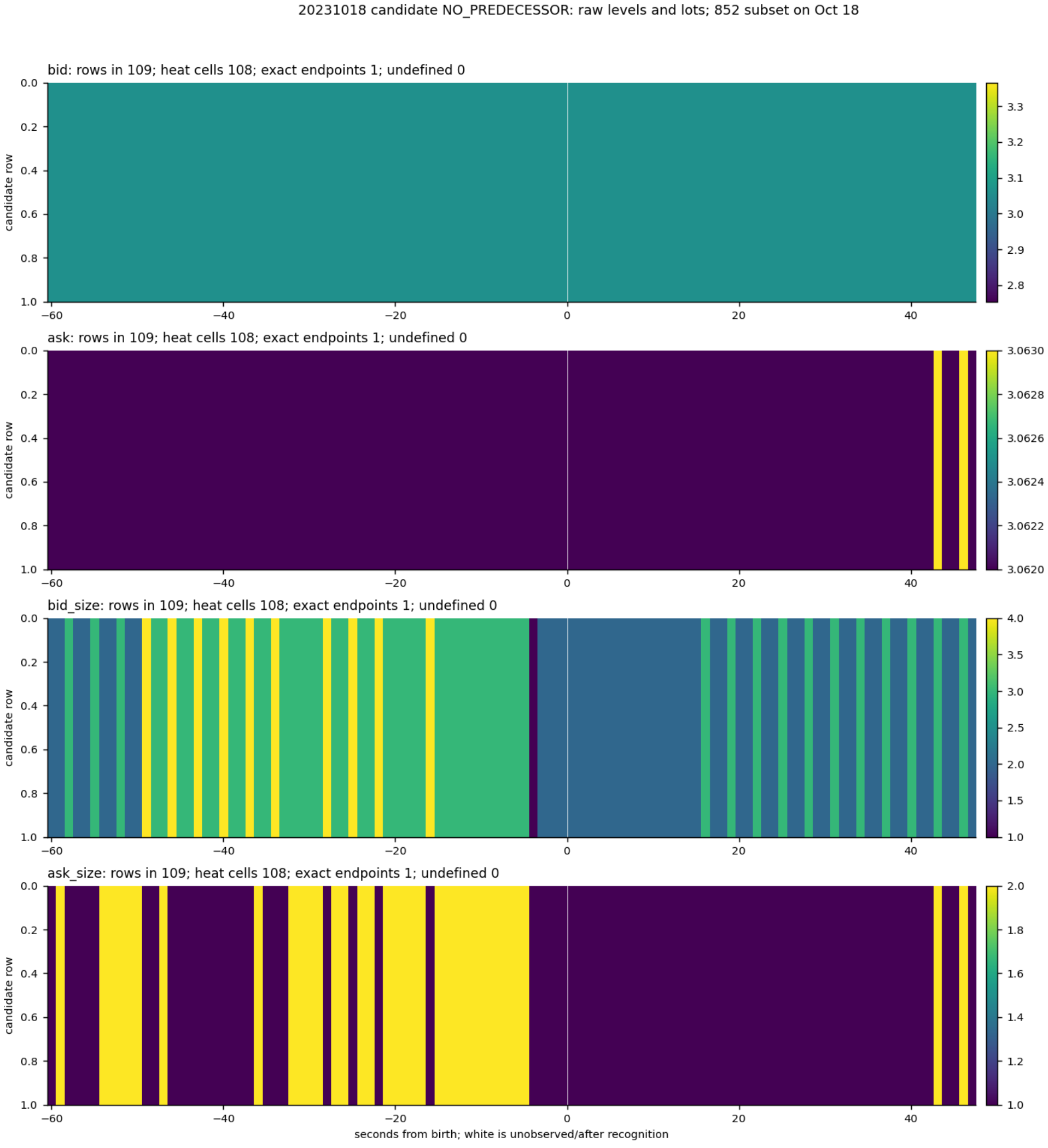
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



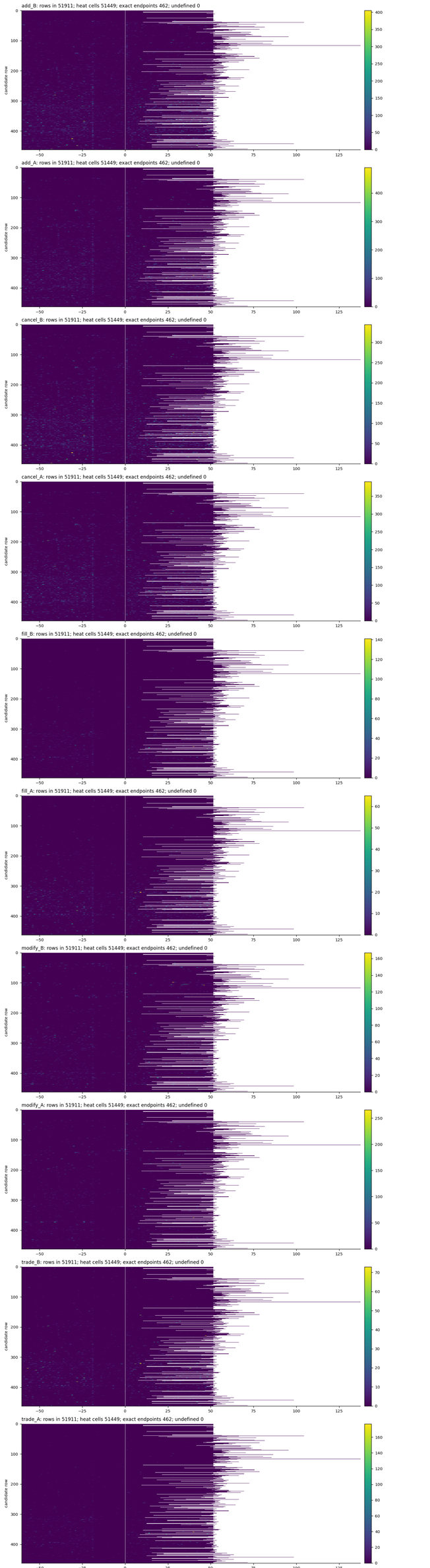
20231018-candidate-NO_PREDECESSOR-flow.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.




20231018 candidate NO_PREDECESSOR: raw levels and lots; 852 subset on Oct 18
bid: rows in 109; heat cells 108; exact endpoints 1; undefined 0
ask: rows in 109; heat cells 108; exact endpoints 1; undefined 0
bid_size: rows in 109; heat cells 108; exact endpoints 1; undefined 0
ask_size: rows in 109; heat cells 108; exact endpoints 1; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

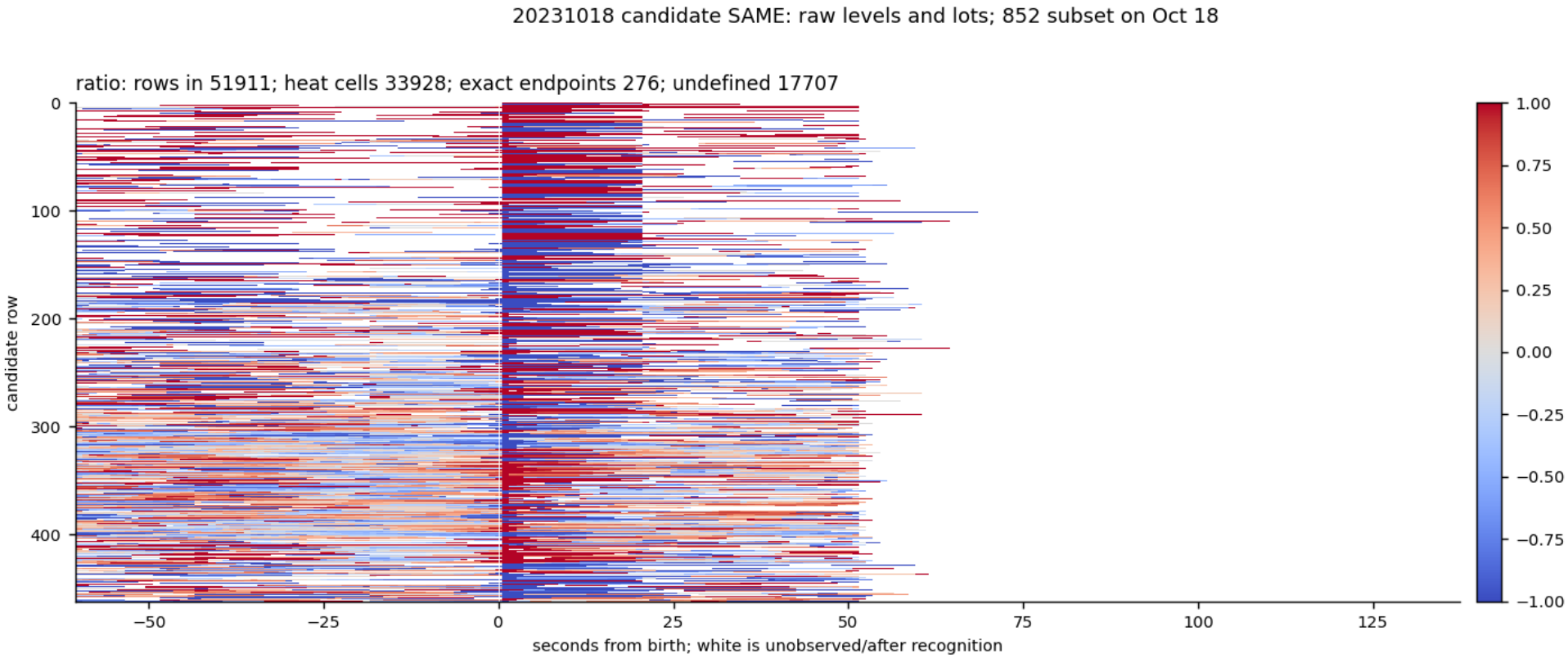
20231018 candidate SAME: raw levels and lots; 852 subset on Oct 18

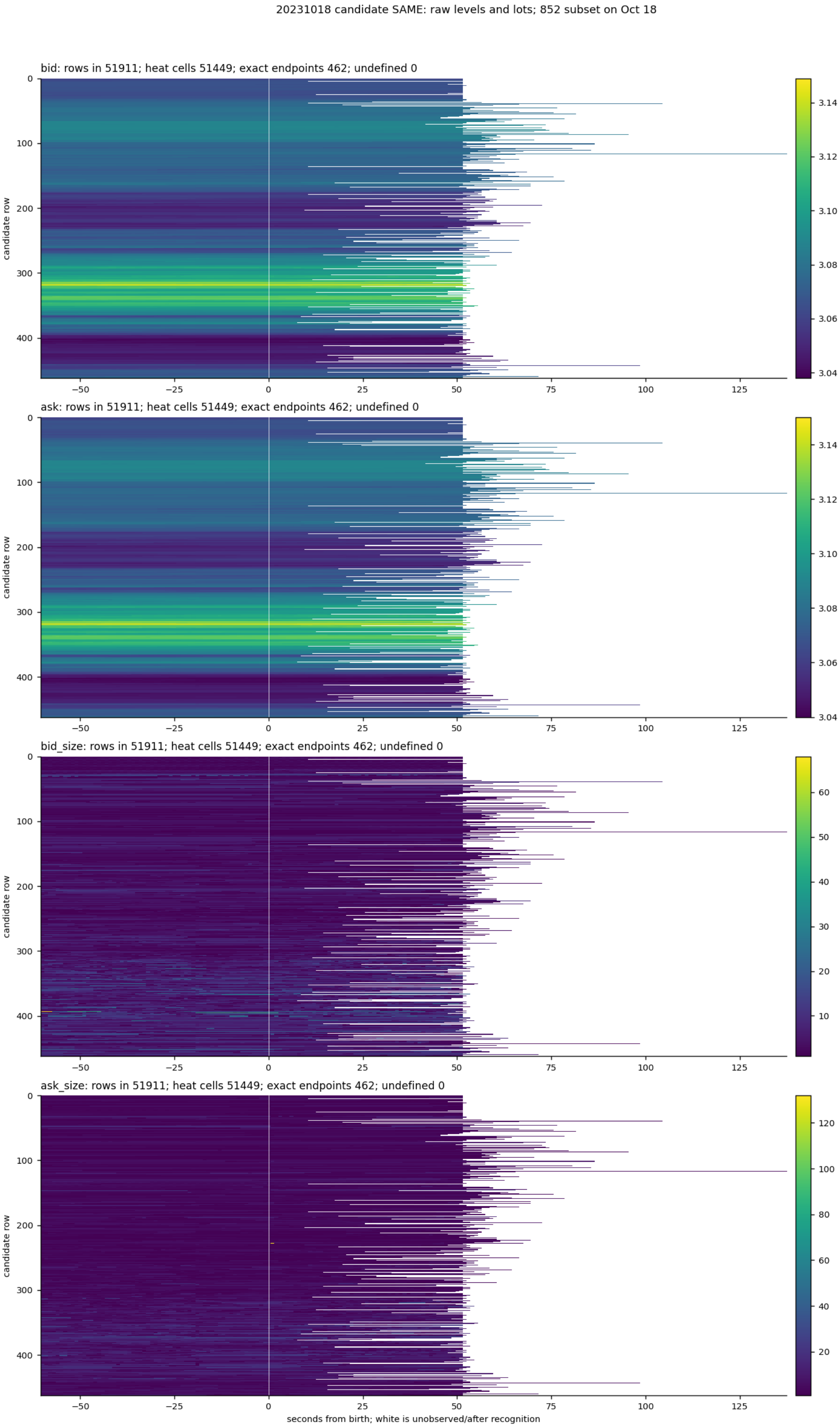


20231018-candidate-SAME-endpoints.png

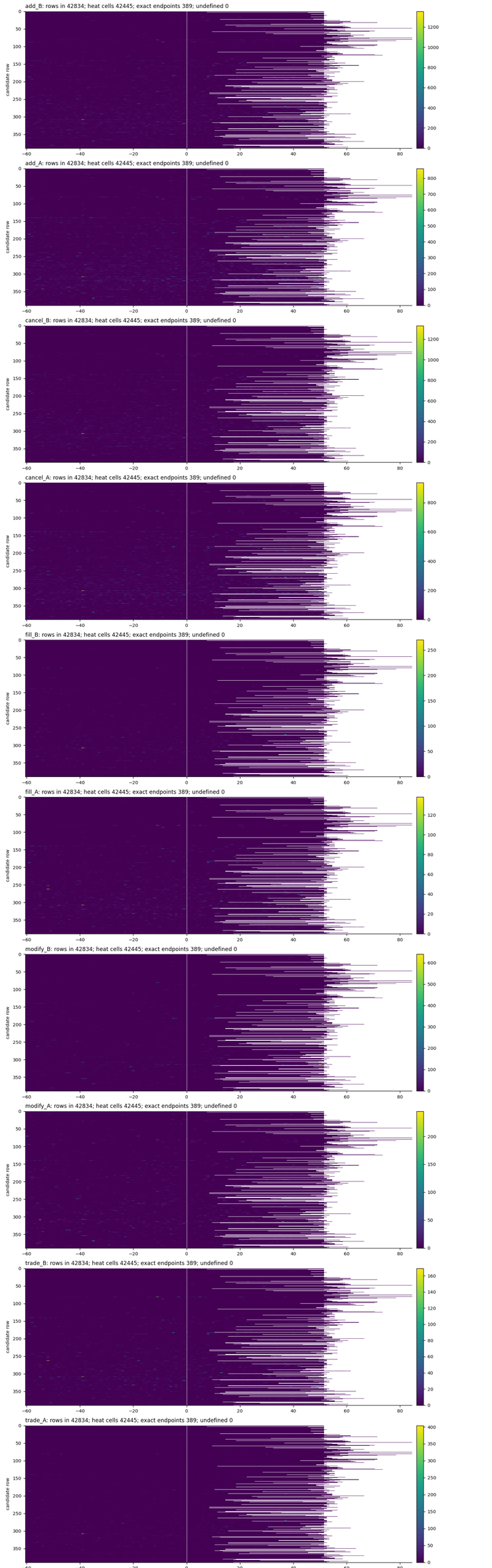
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.




20231018 candidate SAME: raw levels and lots; 852 subset on Oct 18
ratio: rows in 51911; heat cells 33928; exact endpoints 276; undefined 17707
candidate row
0
100
200
300
400
−50
−25
0
25
50
75
100
125
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

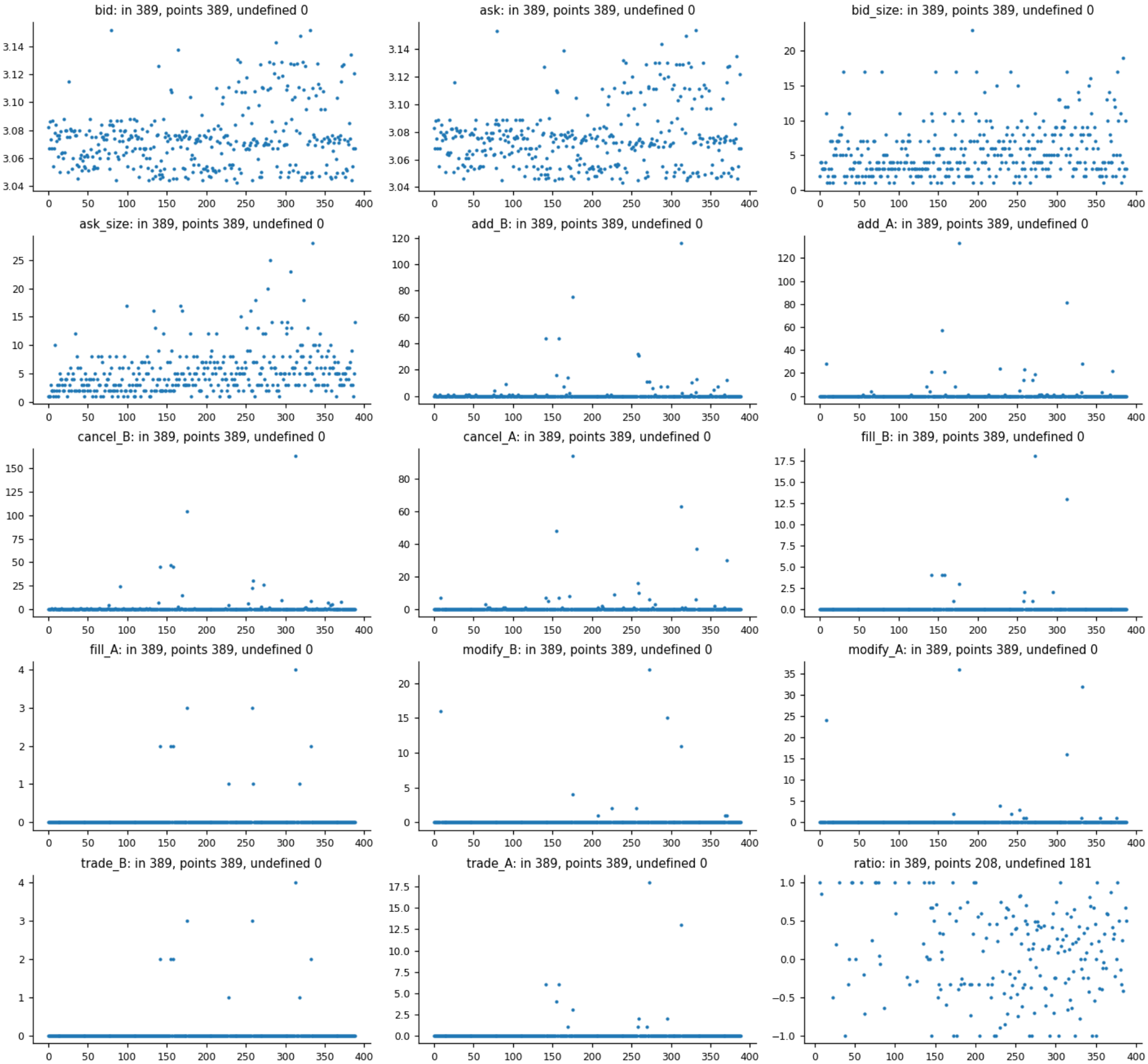

20231018 candidate SAME: raw levels and lots; 852 subset on Oct 18
bid: rows in 51911; heat cells 51449; exact endpoints 462; undefined 0
ask: rows in 51911; heat cells 51449; exact endpoints 462; undefined 0
bid_size: rows in 51911; heat cells 51449; exact endpoints 462; undefined 0
ask_size: rows in 51911; heat cells 51449; exact endpoints 462; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

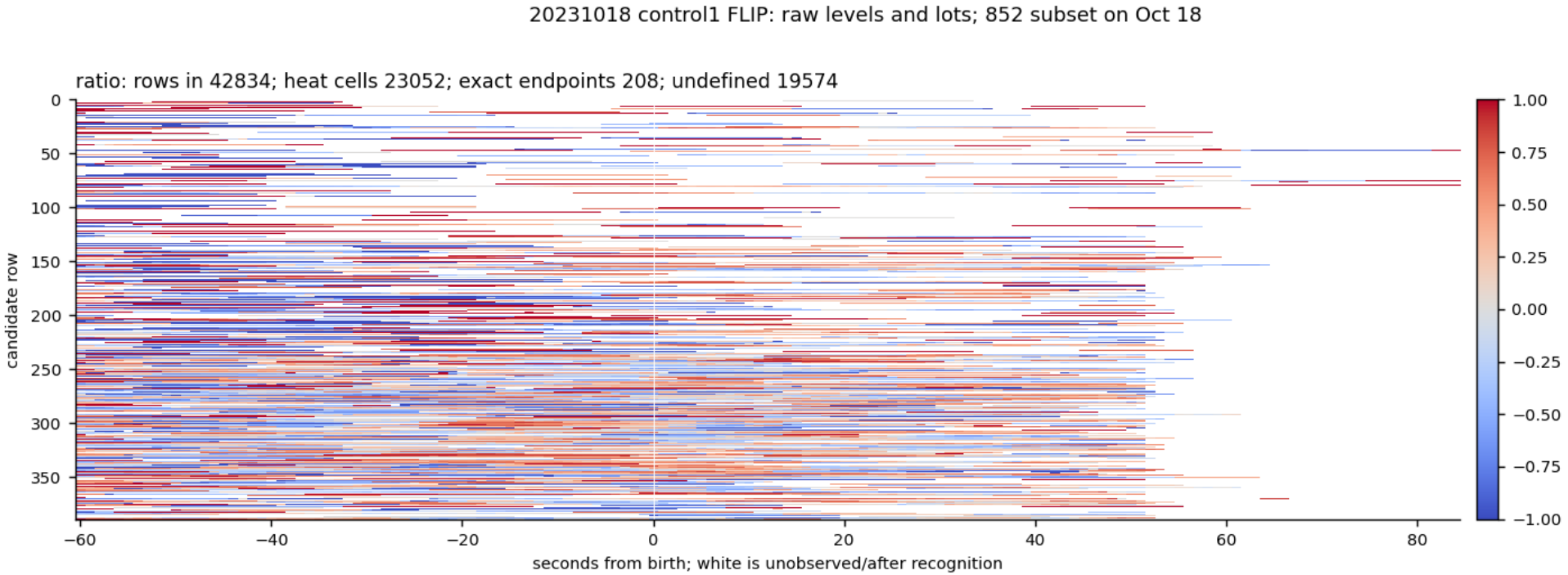
20231018 control1 FLIP: raw levels and lots; 852 subset on Oct 18

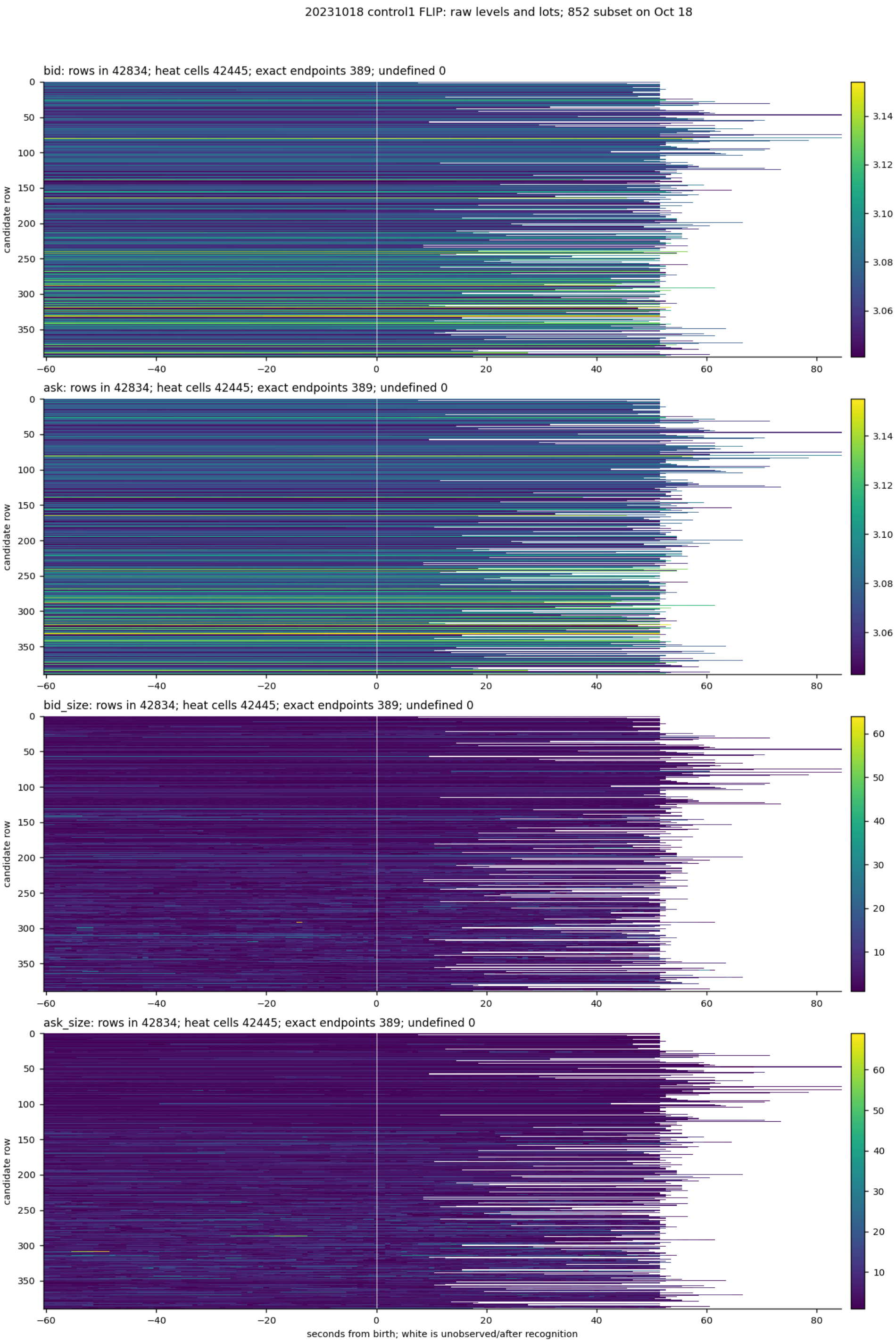


20231018-control1-FLIP-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

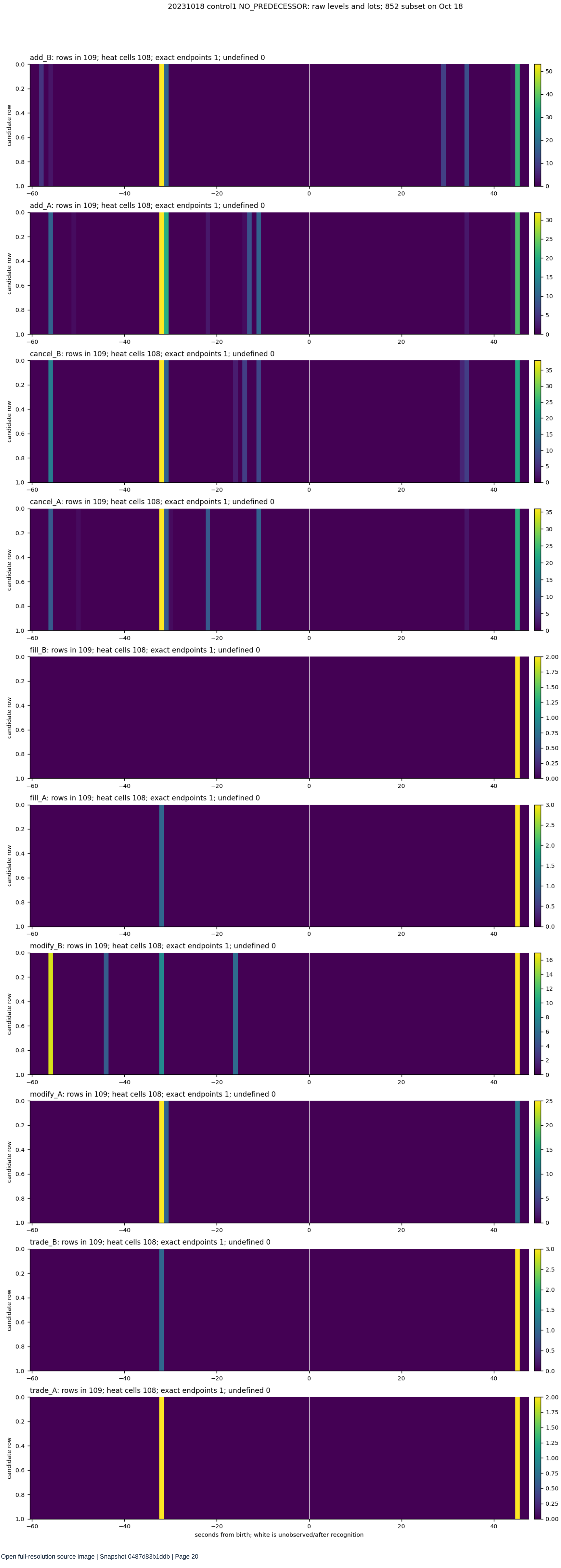



20231018 control1 FLIP: raw levels and lots; 852 subset on Oct 18
ratio: rows in 42834; heat cells 23052; exact endpoints 208; undefined 19574
candidate row
0
50
100
150
200
250
300
350
−60
−40
−20
0
20
40
60
80
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


20231018 control1 FLIP: raw levels and lots; 852 subset on Oct 18
bid: rows in 42834; heat cells 42445; exact endpoints 389; undefined 0
ask: rows in 42834; heat cells 42445; exact endpoints 389; undefined 0
bid_size: rows in 42834; heat cells 42445; exact endpoints 389; undefined 0
ask_size: rows in 42834; heat cells 42445; exact endpoints 389; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

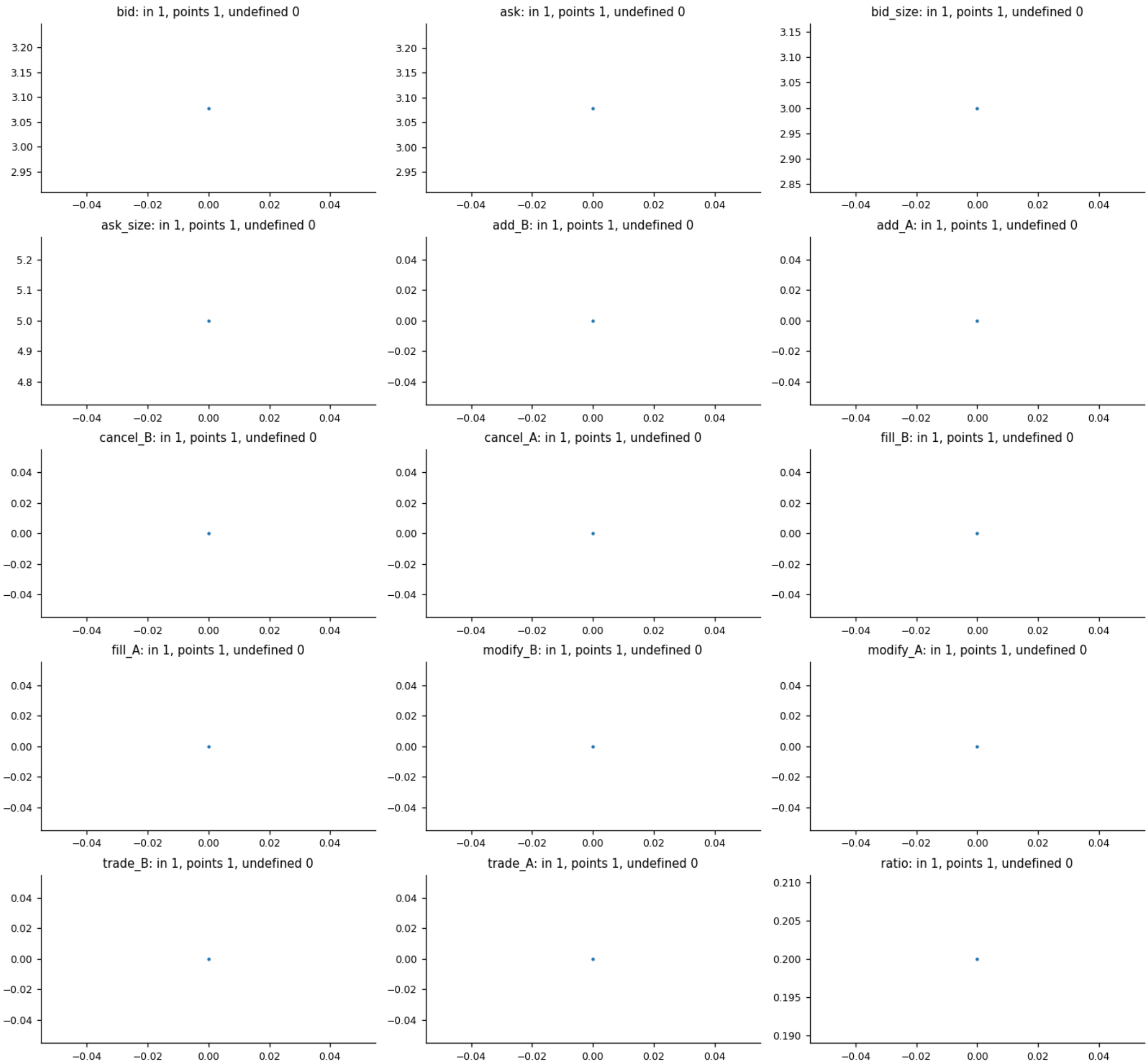
20231018-control1-NO_PREDECESSOR-actions.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

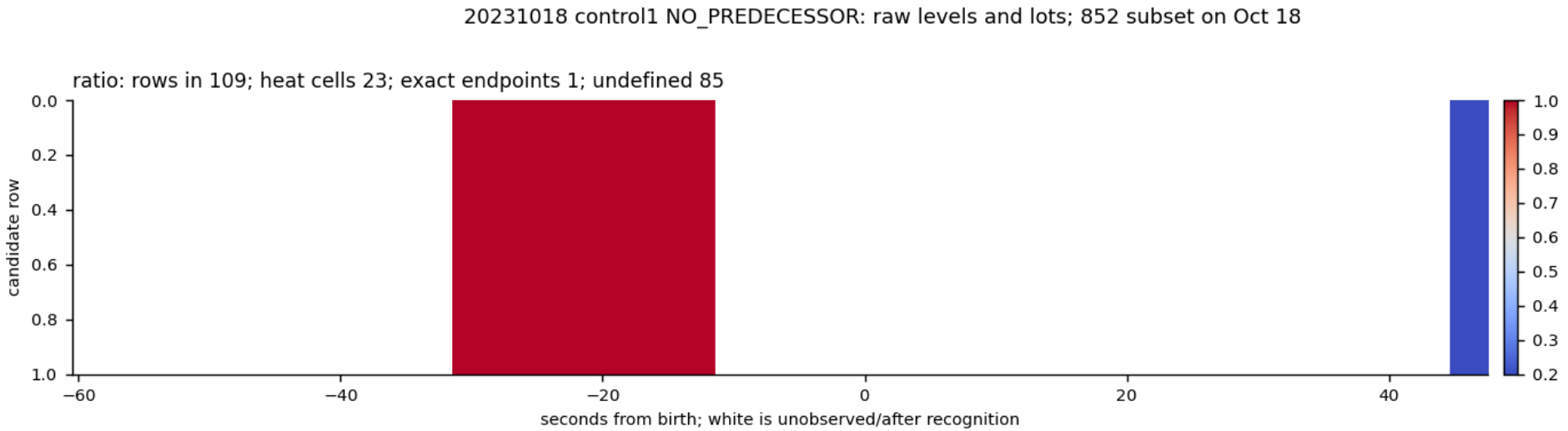


20231018-control1-NO_PREDECESSOR-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

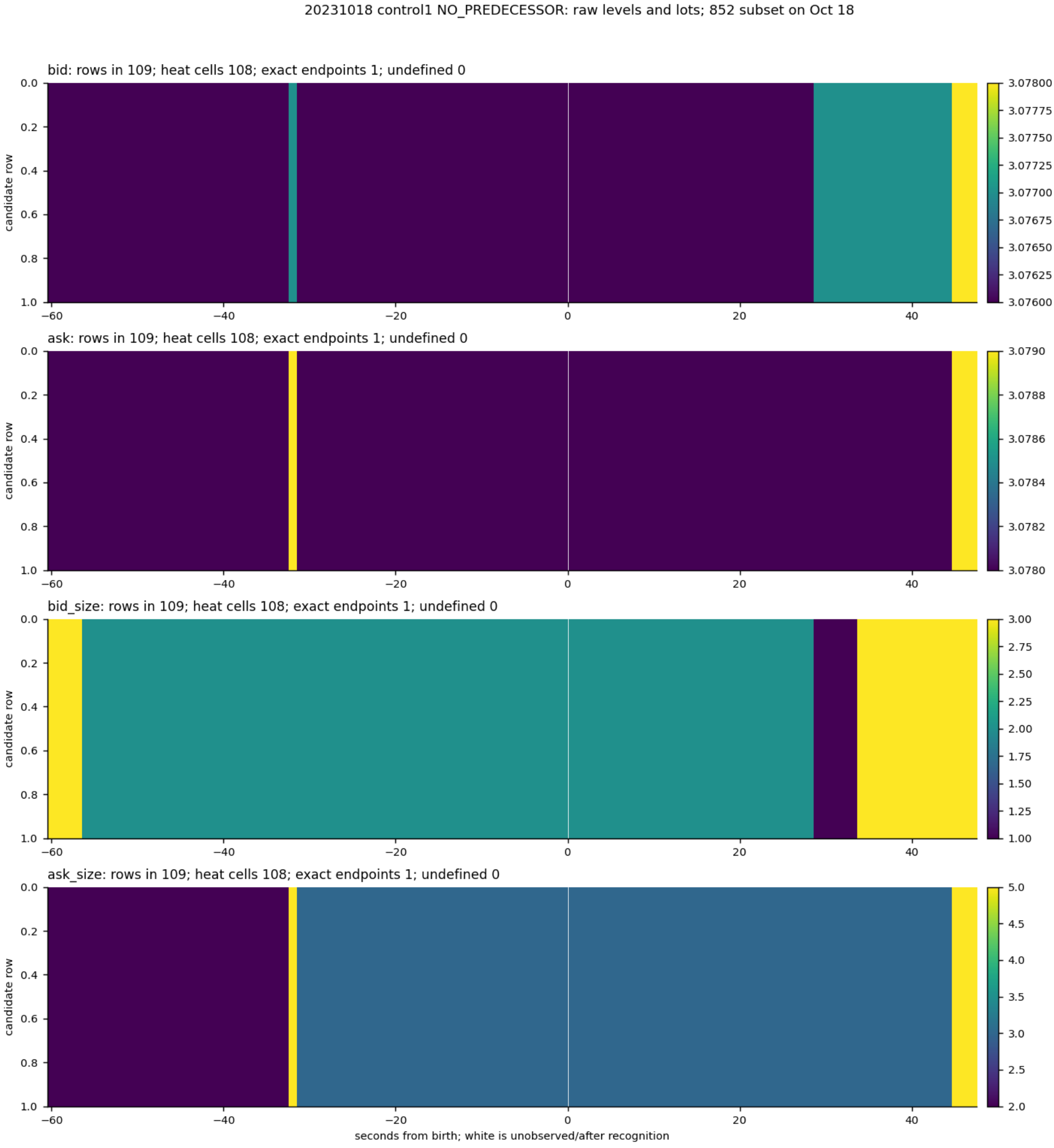


Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

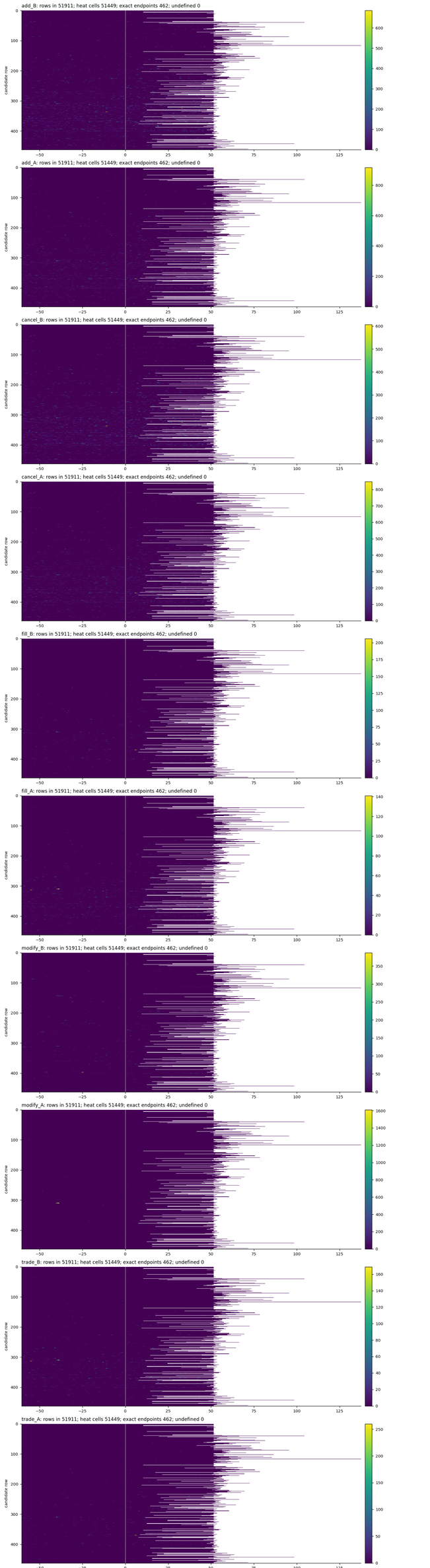


20231018-control1-NO_PREDECESSOR-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

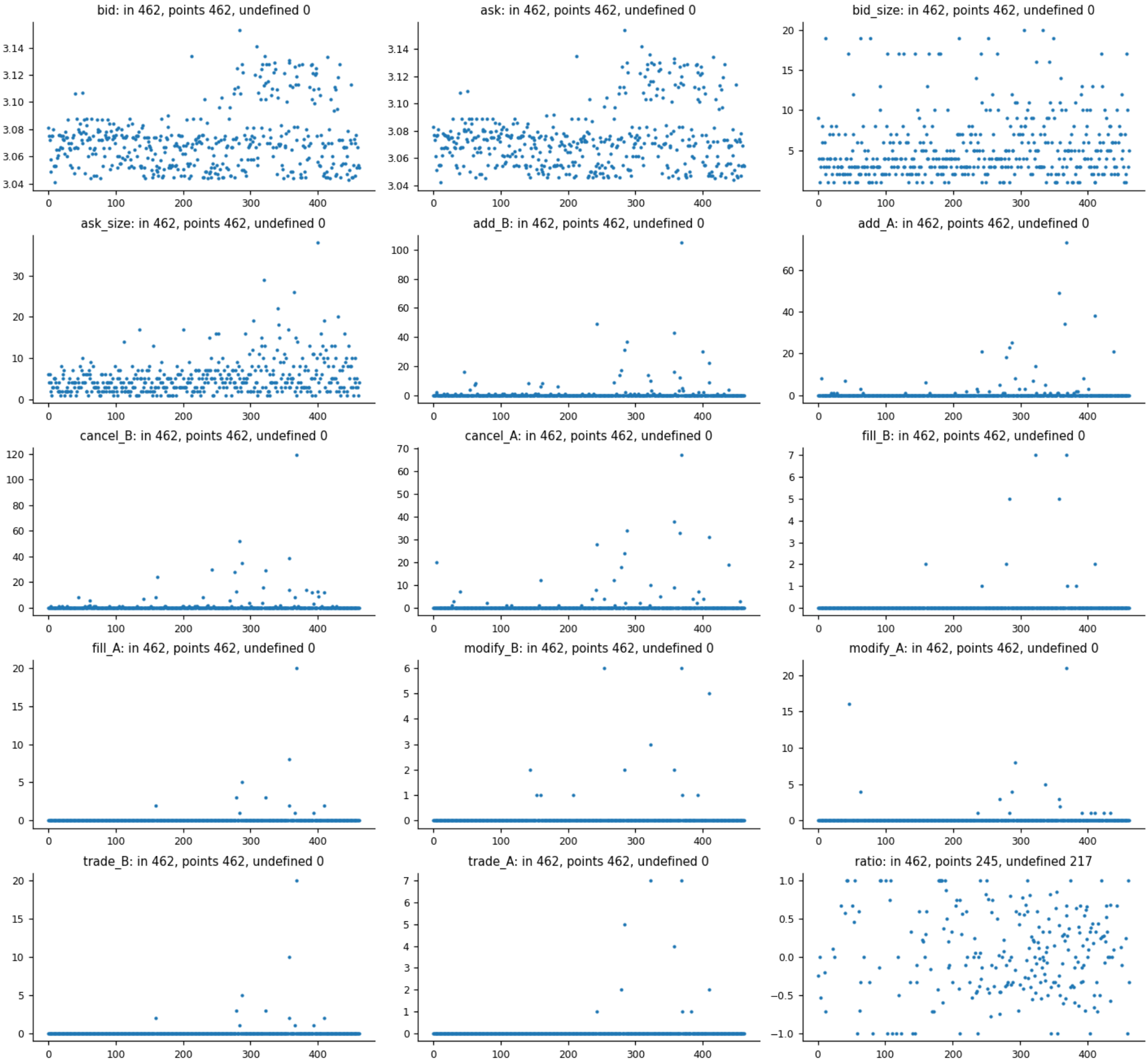


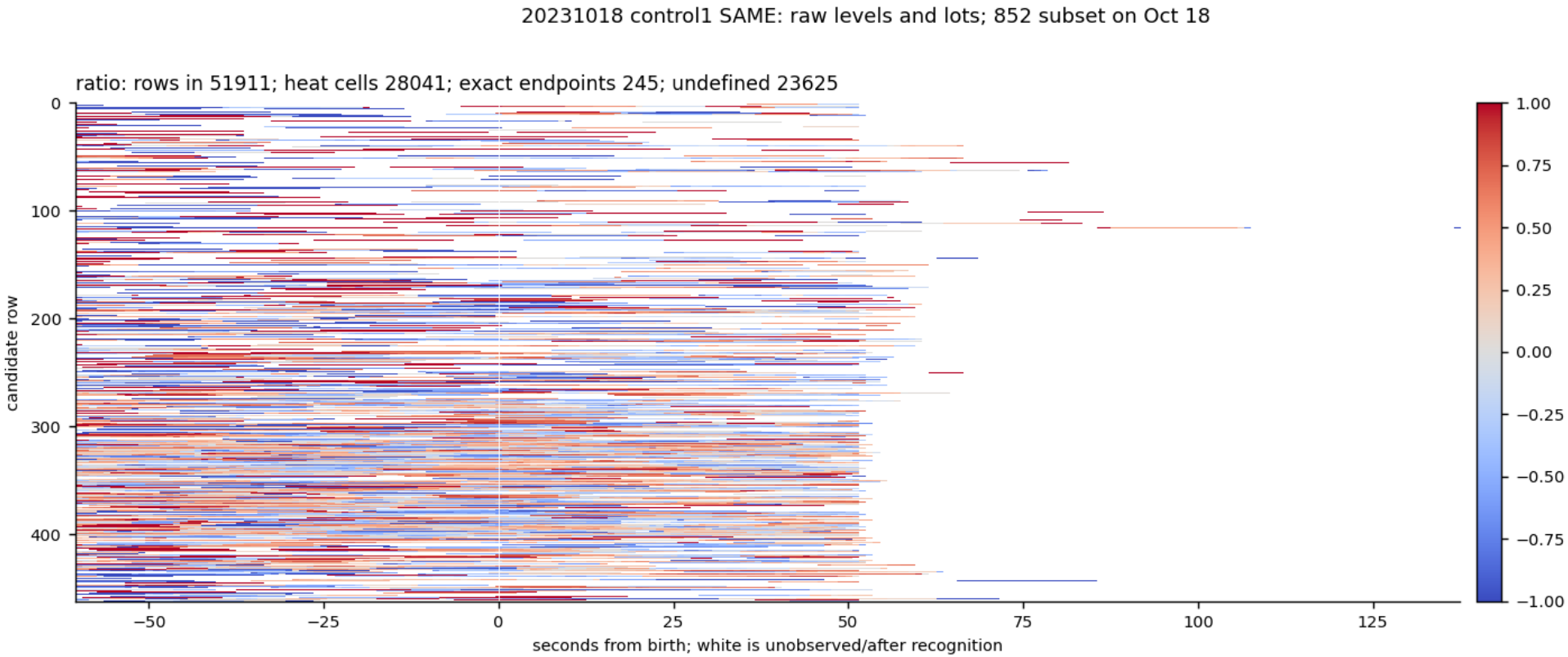
20231018 control1 SAME: raw levels and lots; 852 subset on Oct 18



20231018-control1-SAME-endpoints.png

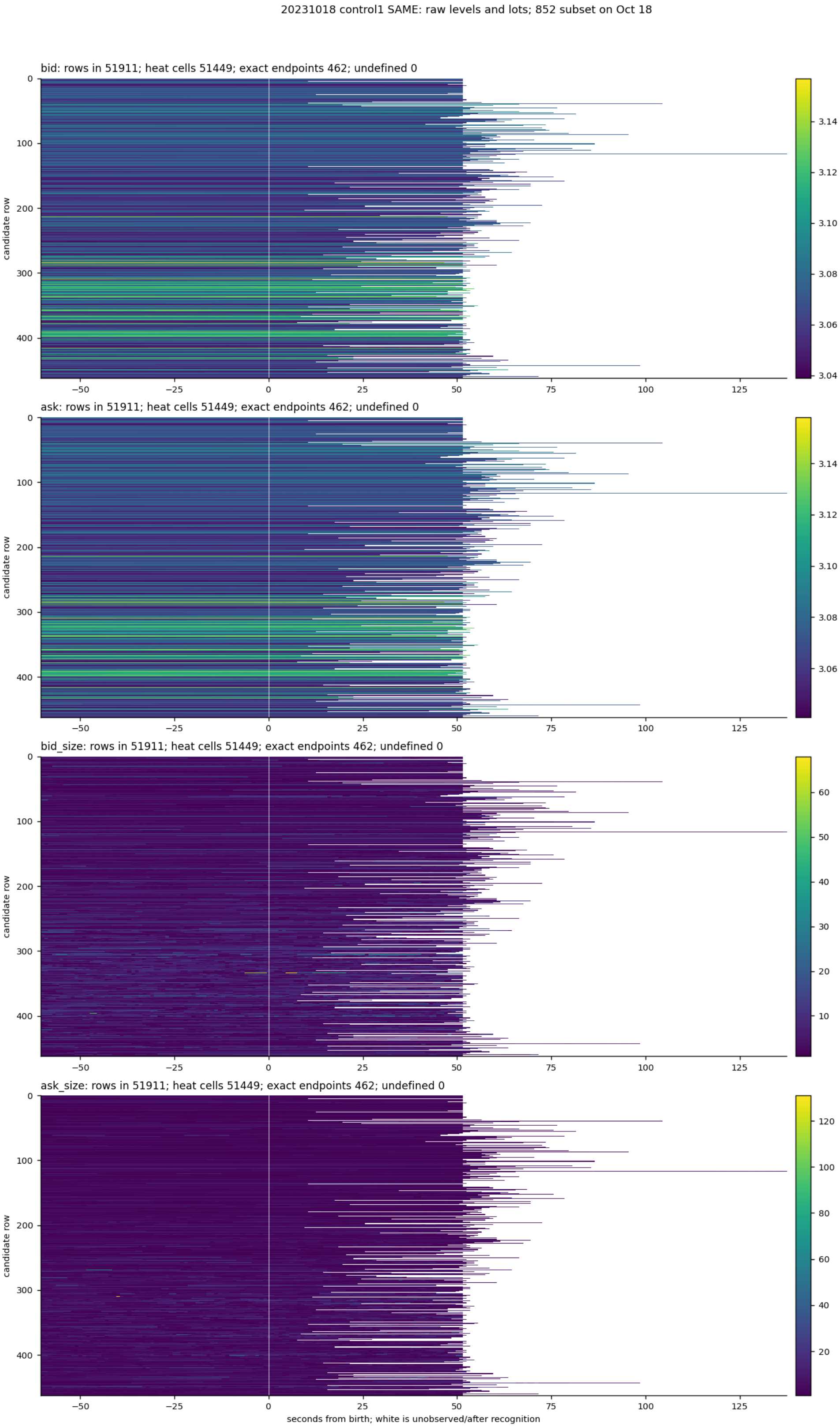
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



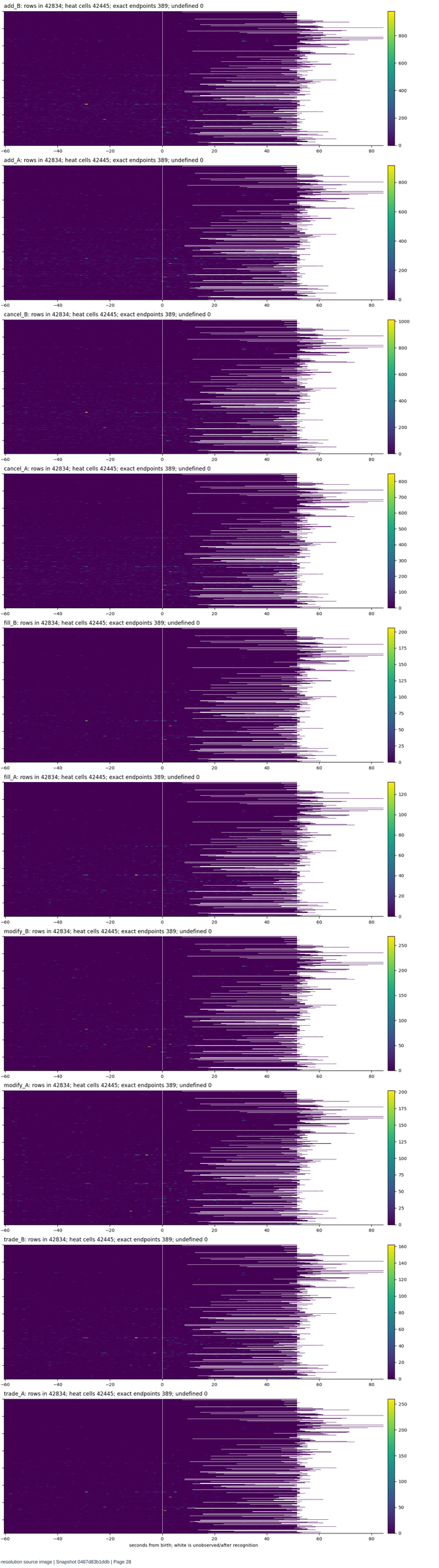

20231018 control1 SAME: raw levels and lots; 852 subset on Oct 18
ratio: rows in 51911; heat cells 28041; exact endpoints 245; undefined 23625
candidate row
0
100
200
300
400
−50
−25
0
25
50
75
100
125
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

20231018-control1-SAME-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

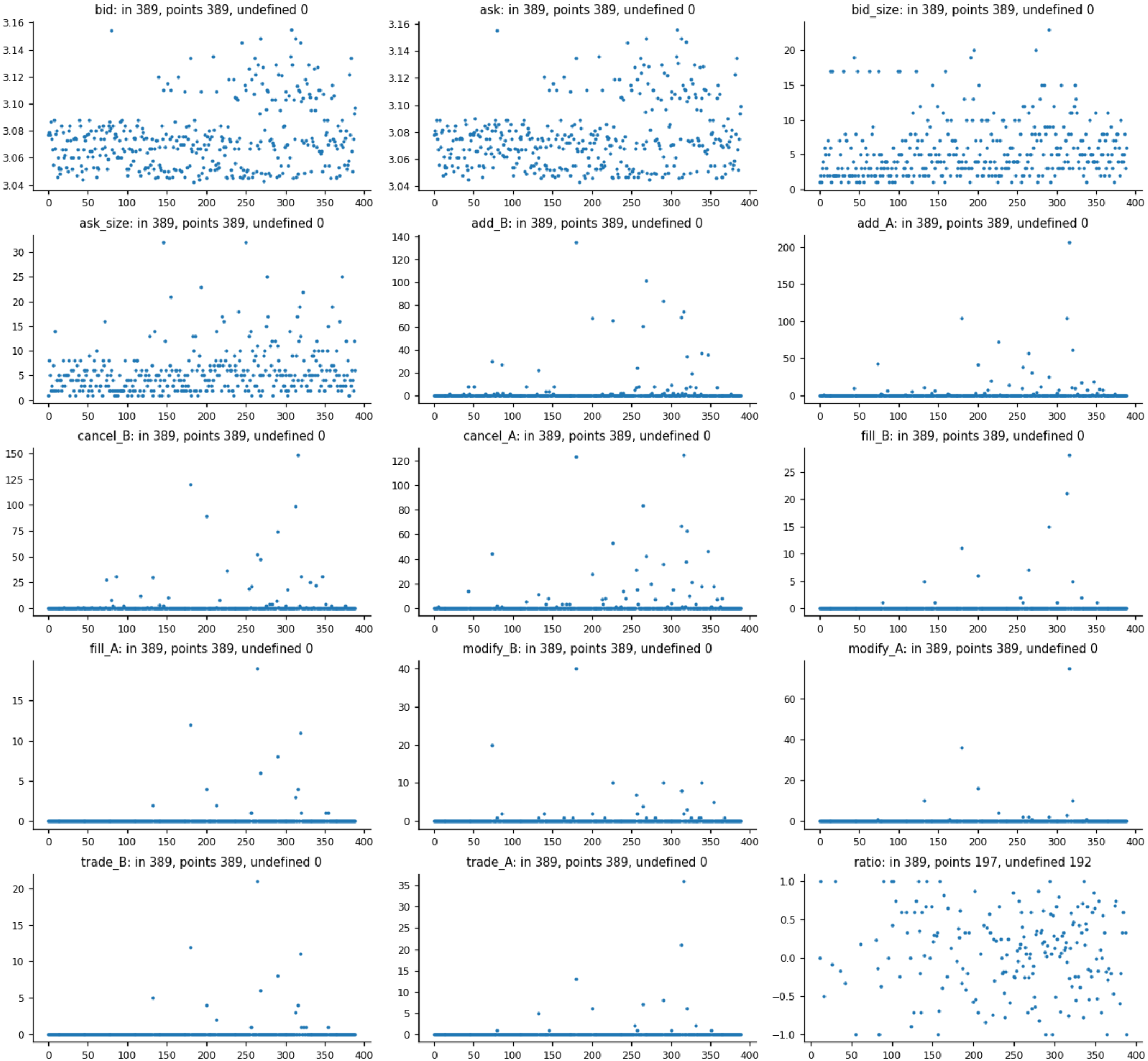


20231018 control2 FLIP: raw levels and lots; 852 subset on Oct 18



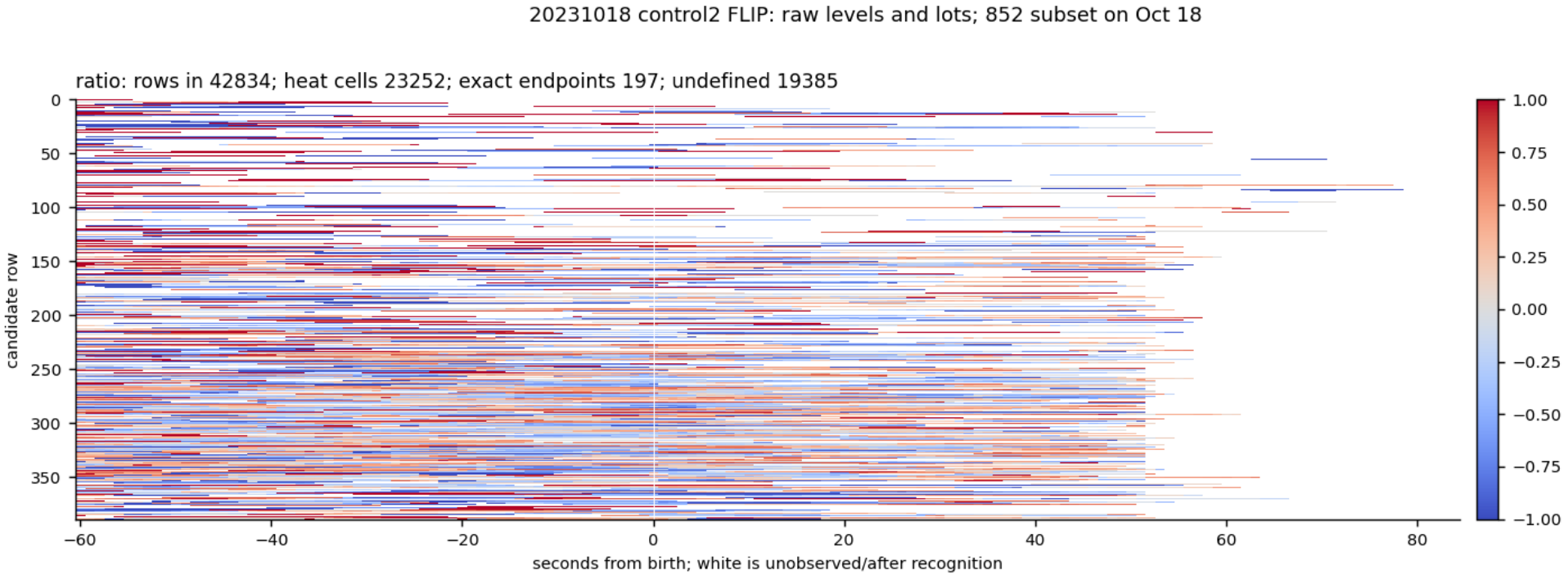
20231018-control2-FLIP-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



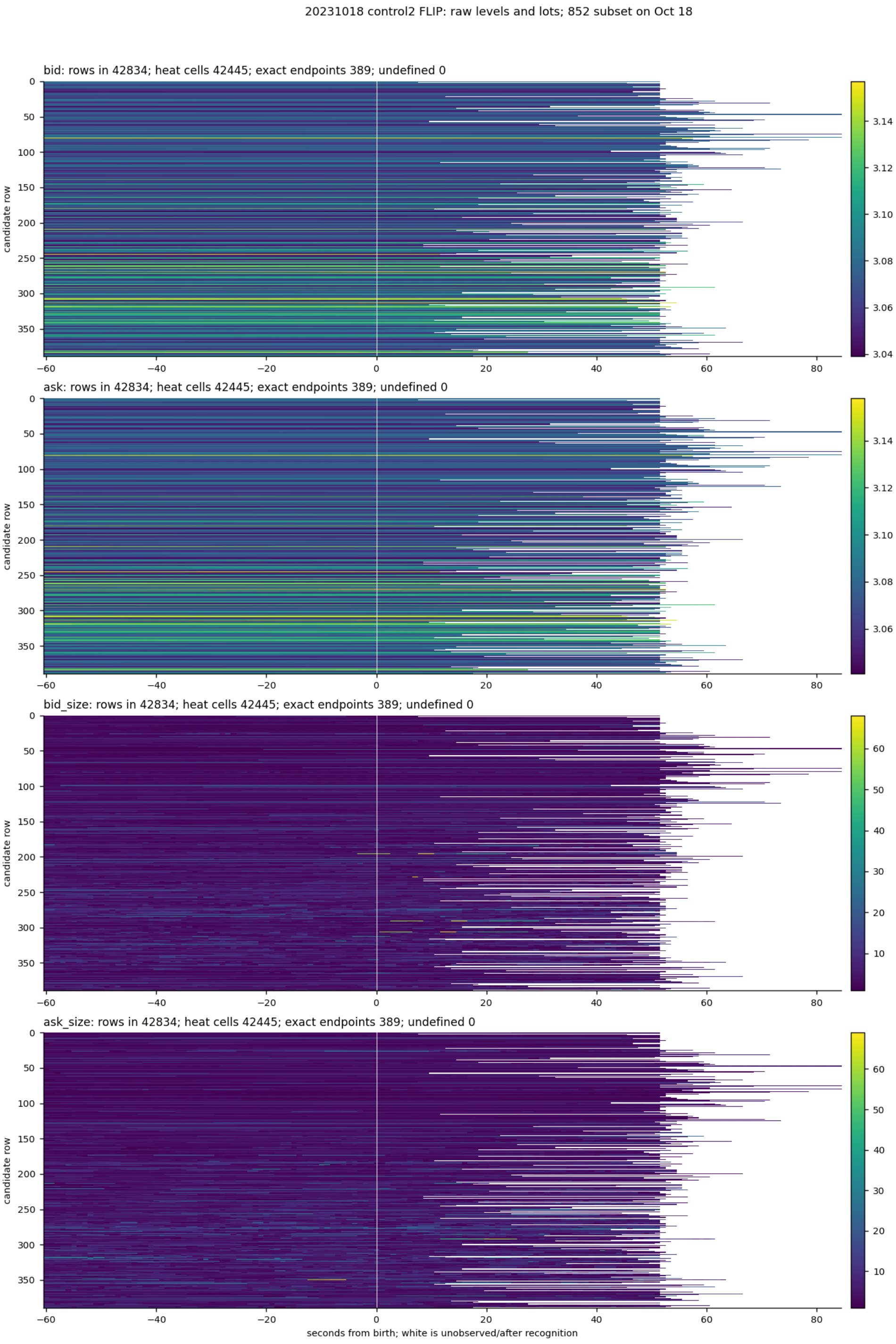
20231018-control2-FLIP-flow.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



20231018-control2-FLIP-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

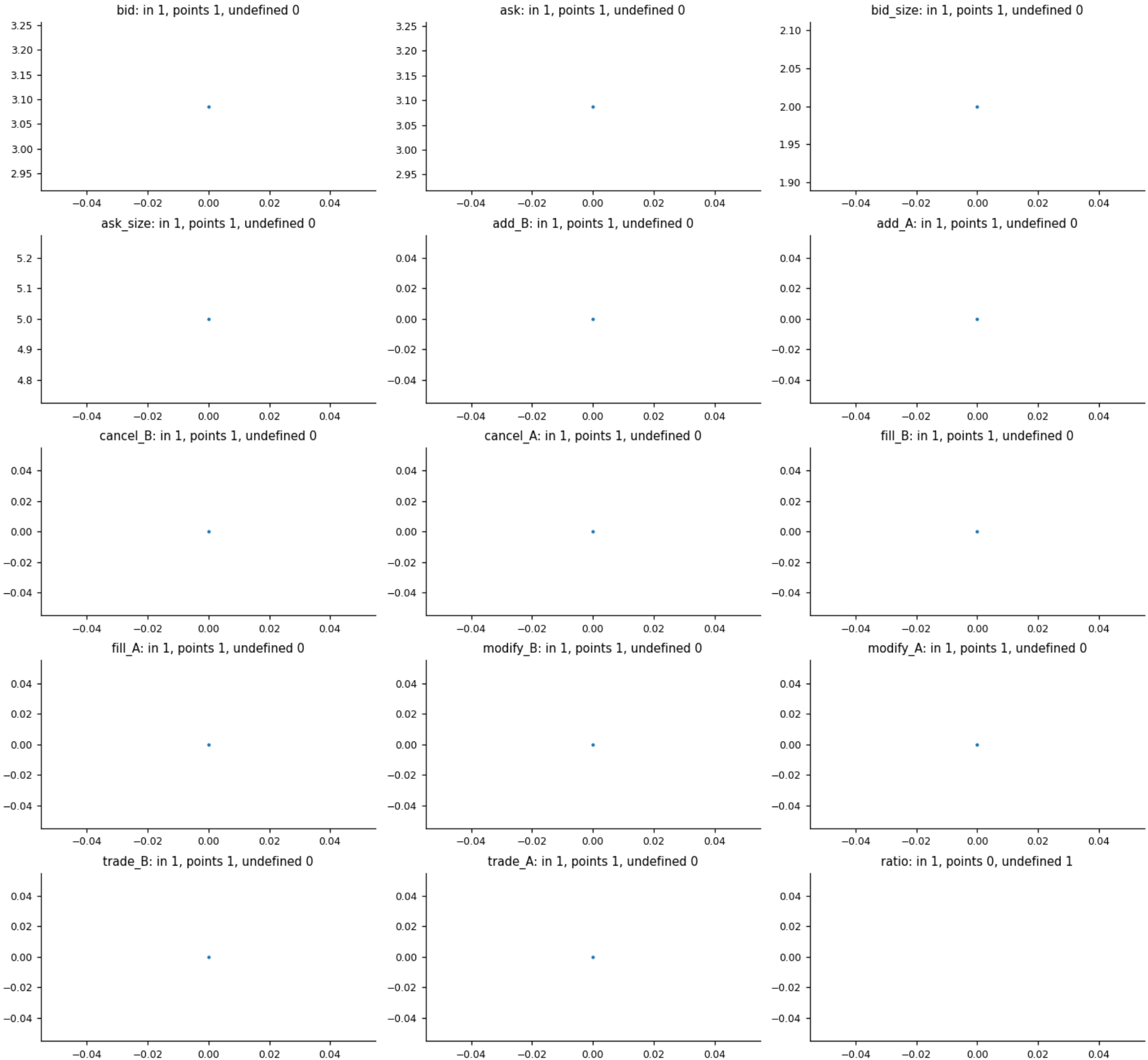


20231018-control2-NO_PREDECESSOR-actions.png

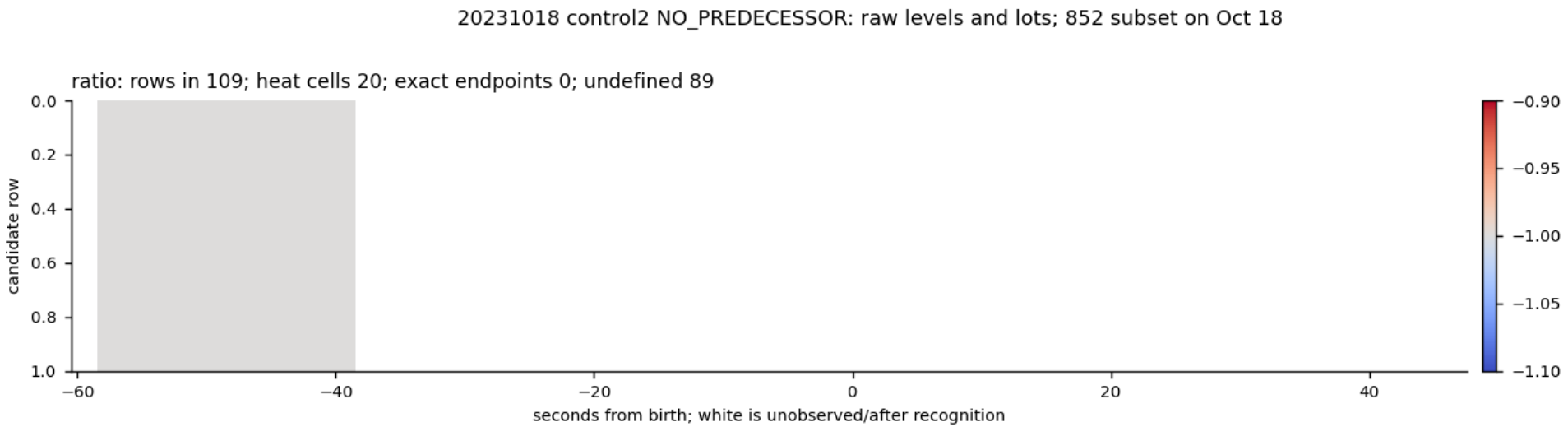
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

20231018-control2-NO_PREDECESSOR-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

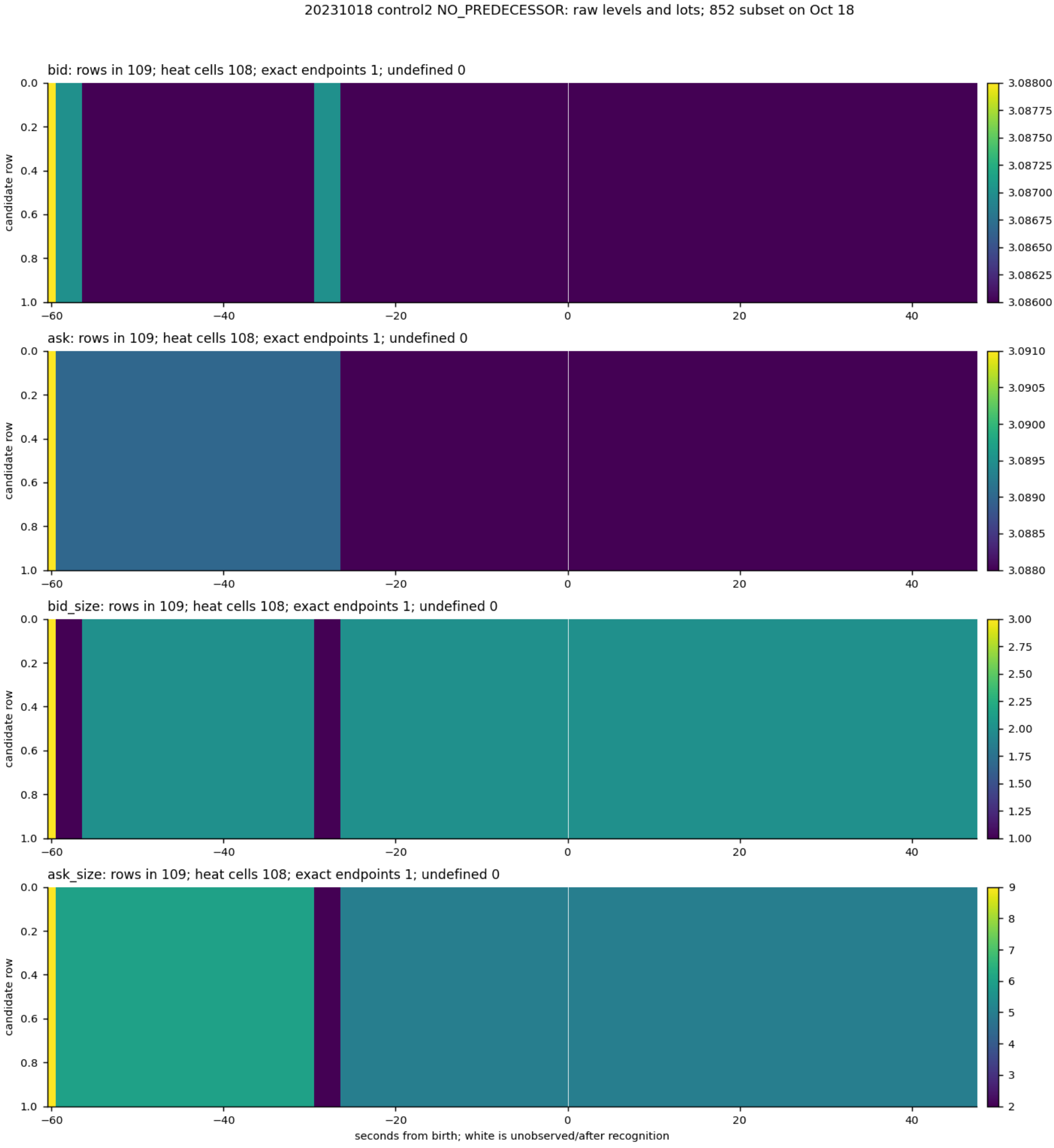


Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

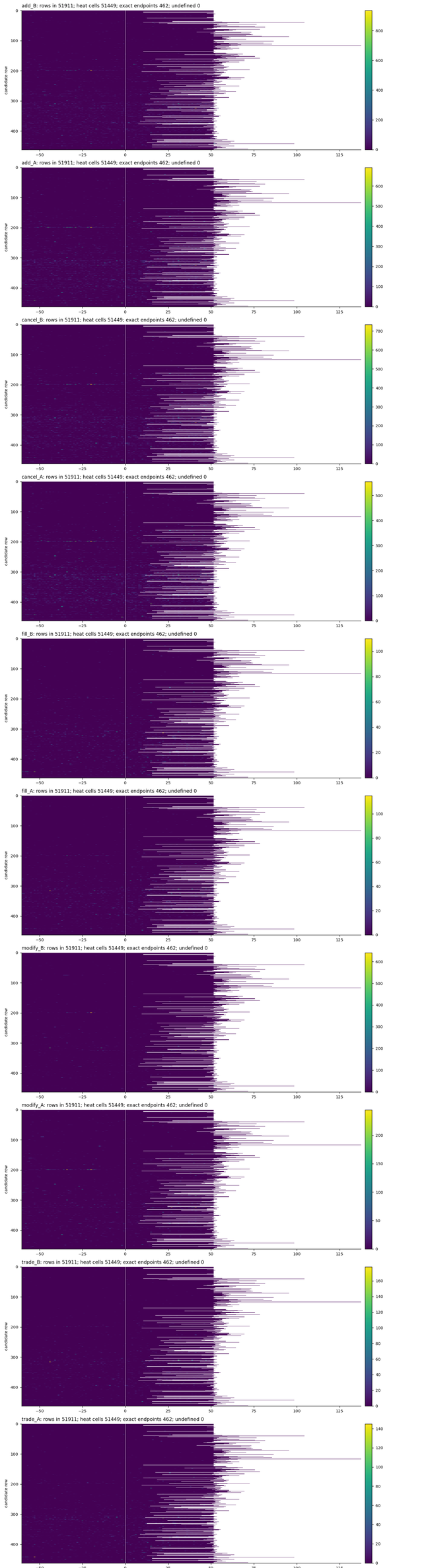


20231018-control2-NO_PREDECESSOR-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

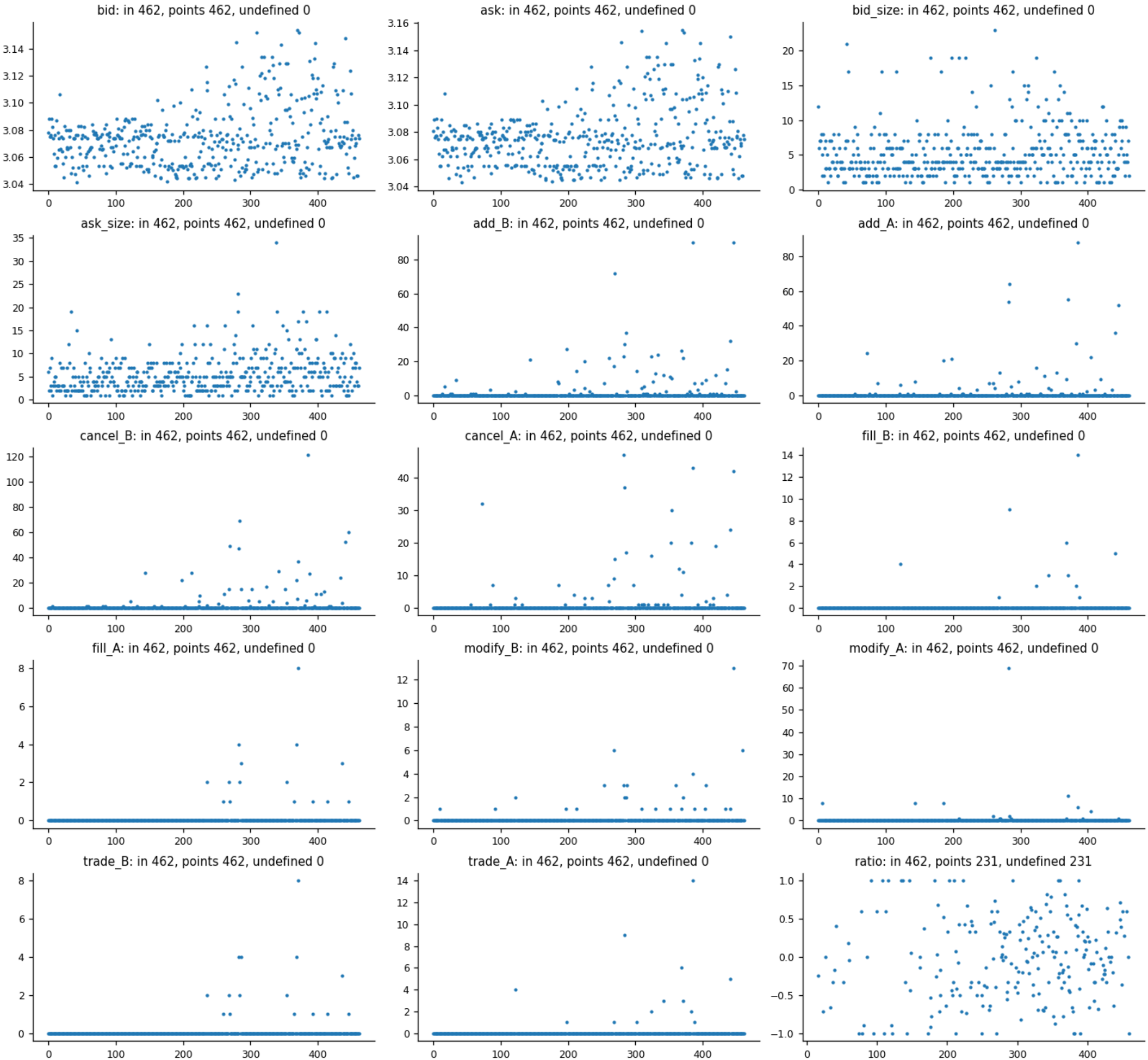


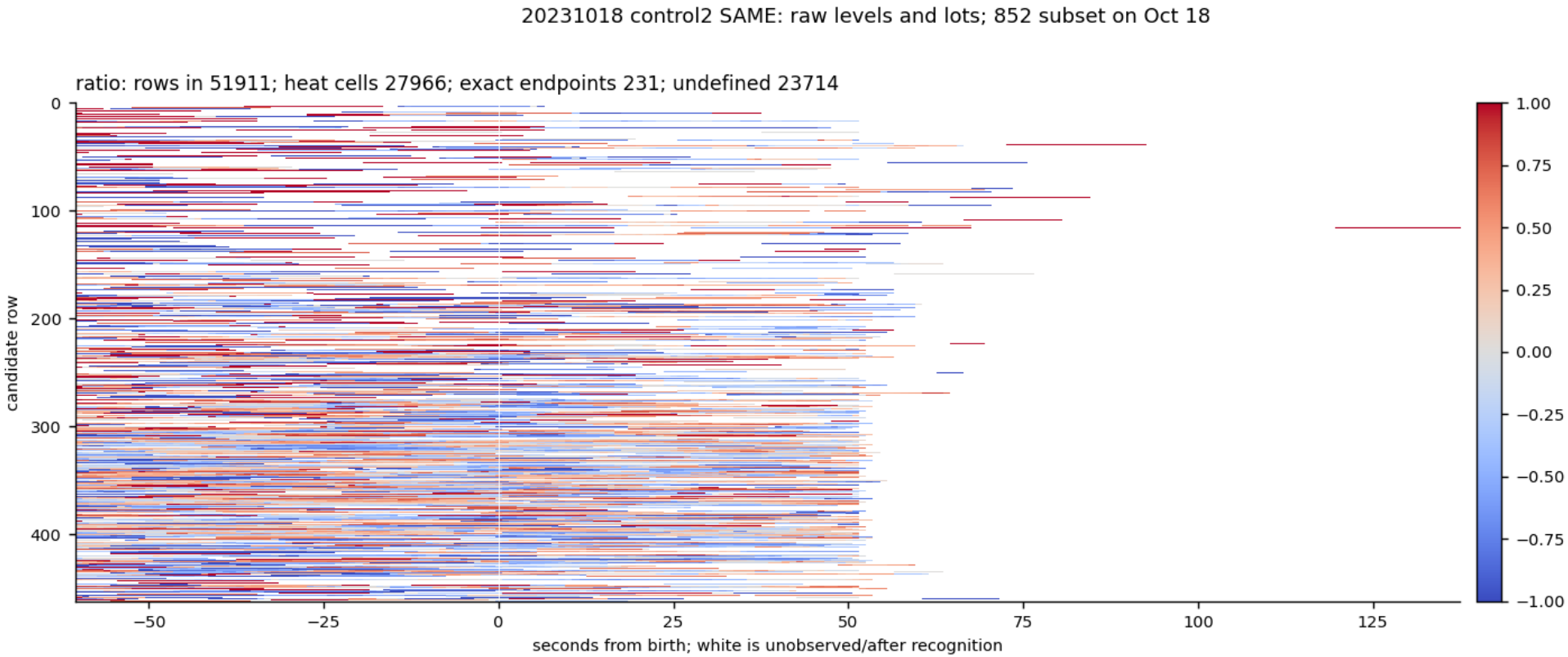
20231018 control2 SAME: raw levels and lots; 852 subset on Oct 18



20231018-control2-SAME-endpoints.png

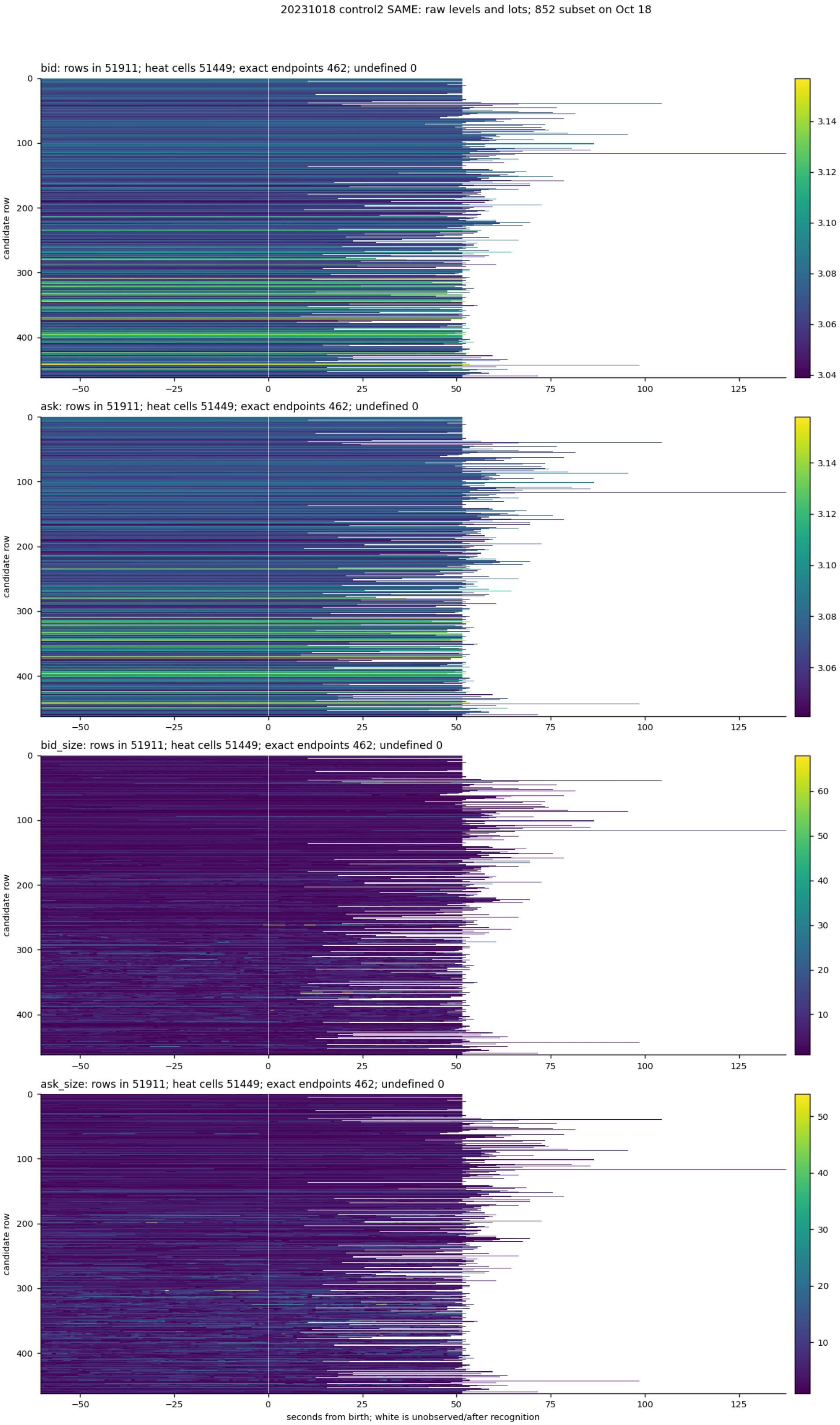
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

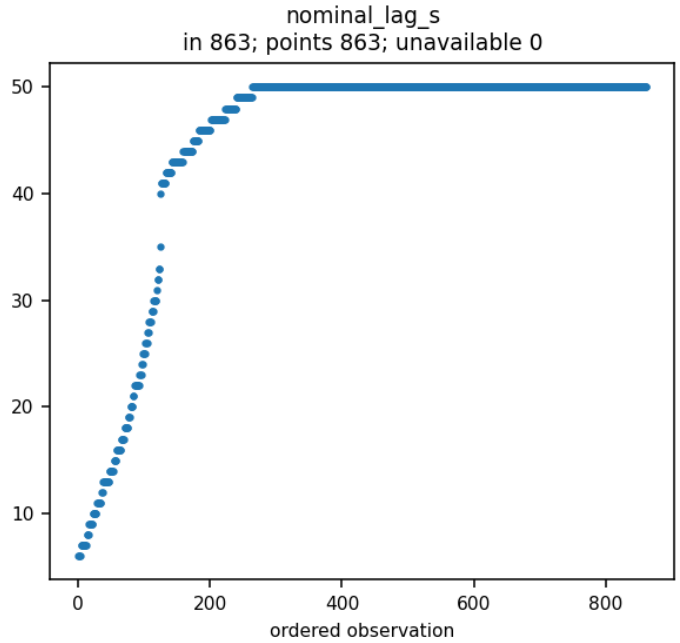


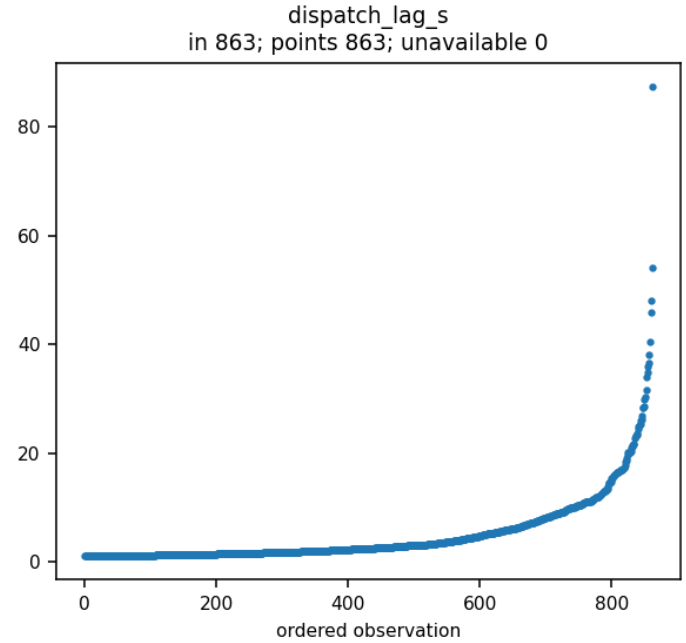

20231018 control2 SAME: raw levels and lots; 852 subset on Oct 18
ratio: rows in 51911; heat cells 27966; exact endpoints 231; undefined 23714
candidate row
0
100
200
300
400
−50
−25
0
25
50
75
100
125
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

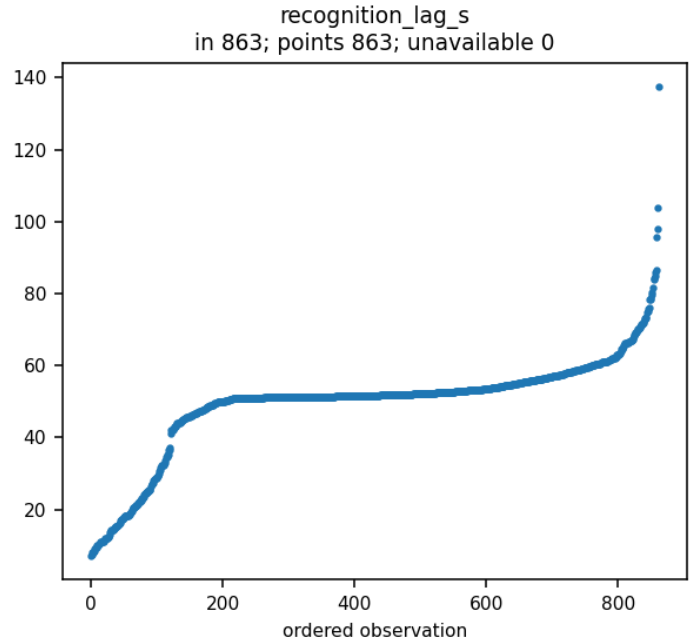
20231018-control2-SAME-quotes.png

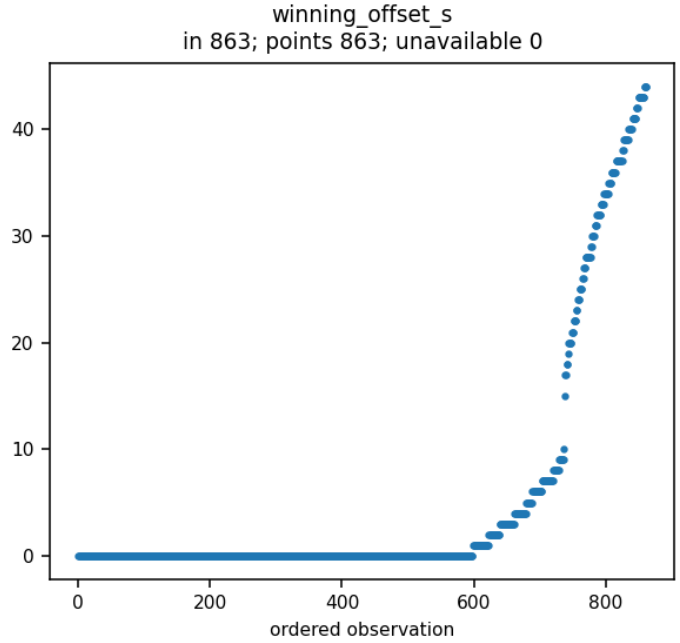
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

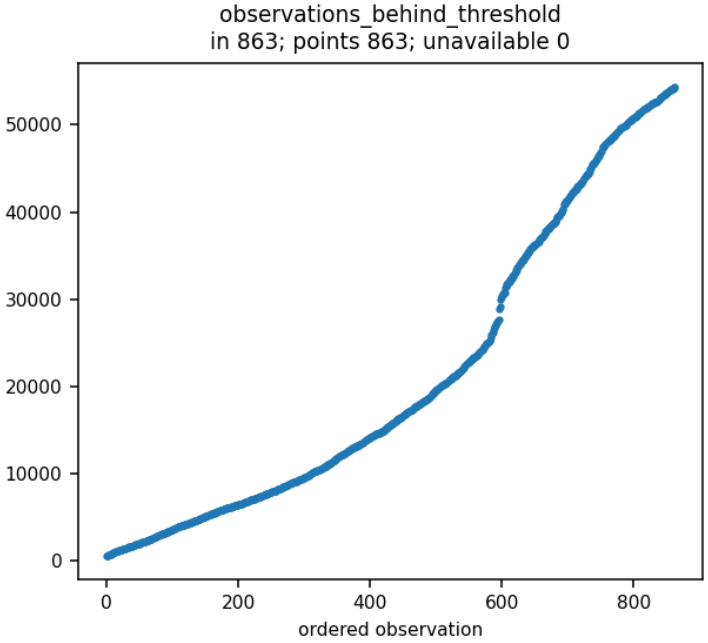


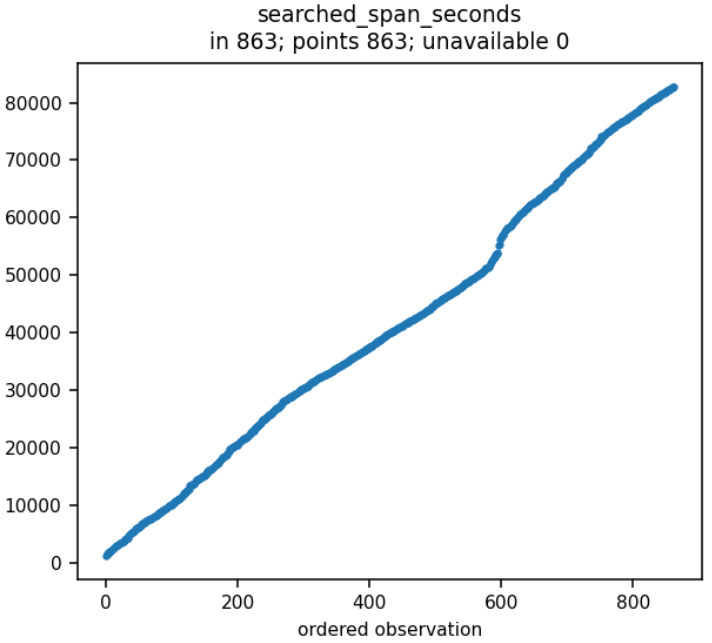

nominal_lag_s
in 863; points 863; unavailable 0
ordered observation

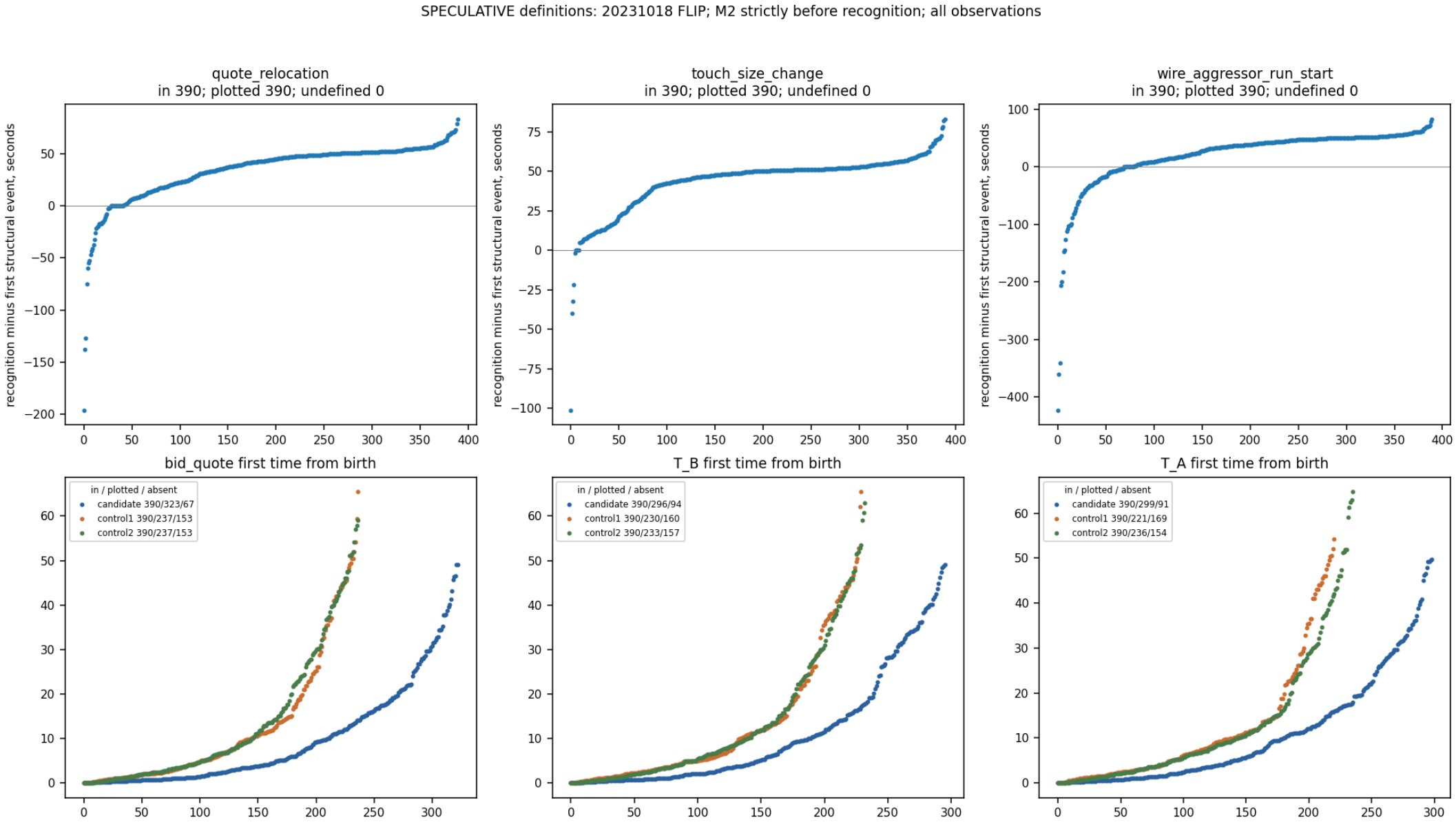

dispatch_lag_s
in 863; points 863; unavailable 0
ordered observation

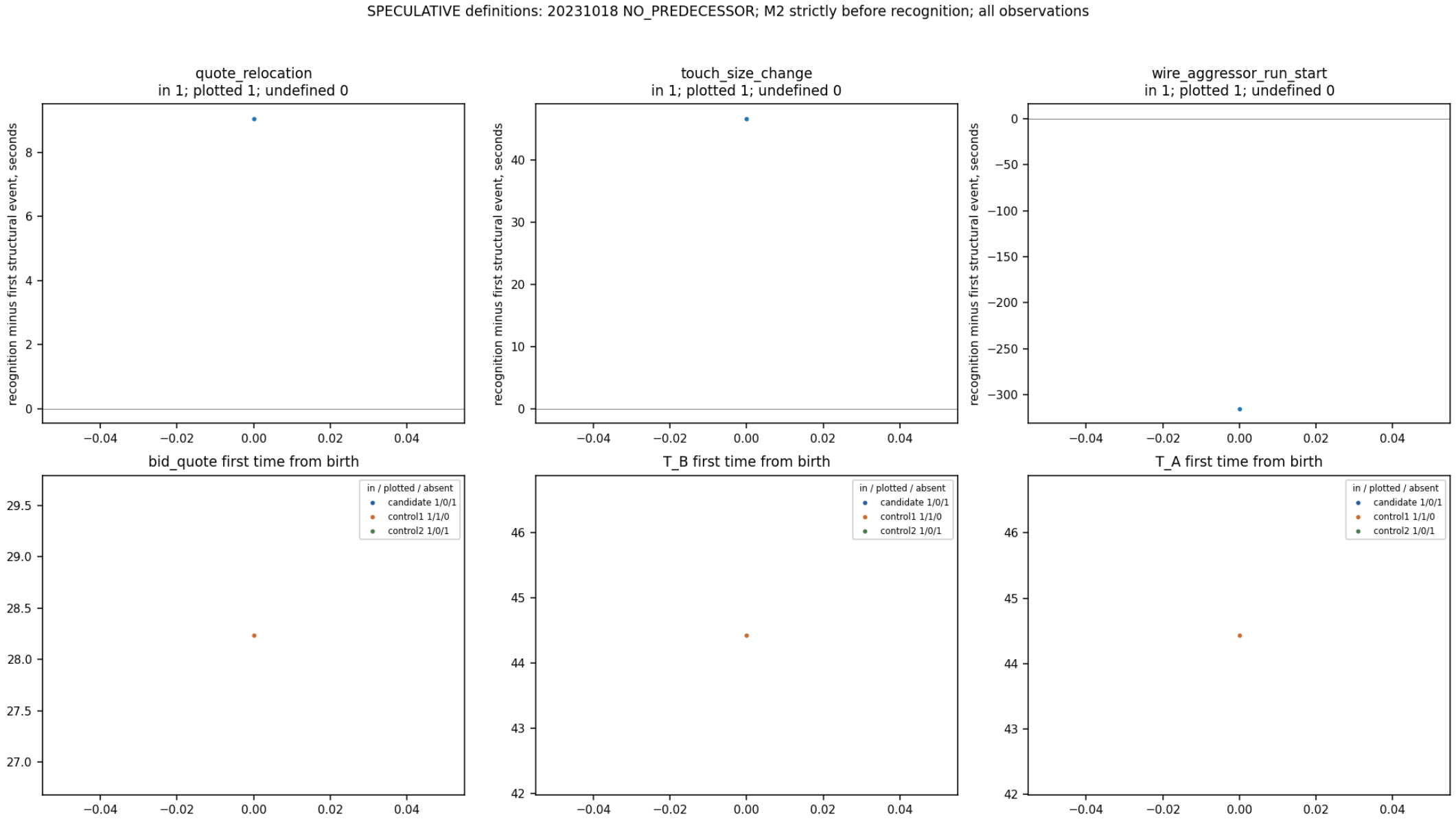

recognition_lag_s
in 863; points 863; unavailable 0
ordered observation

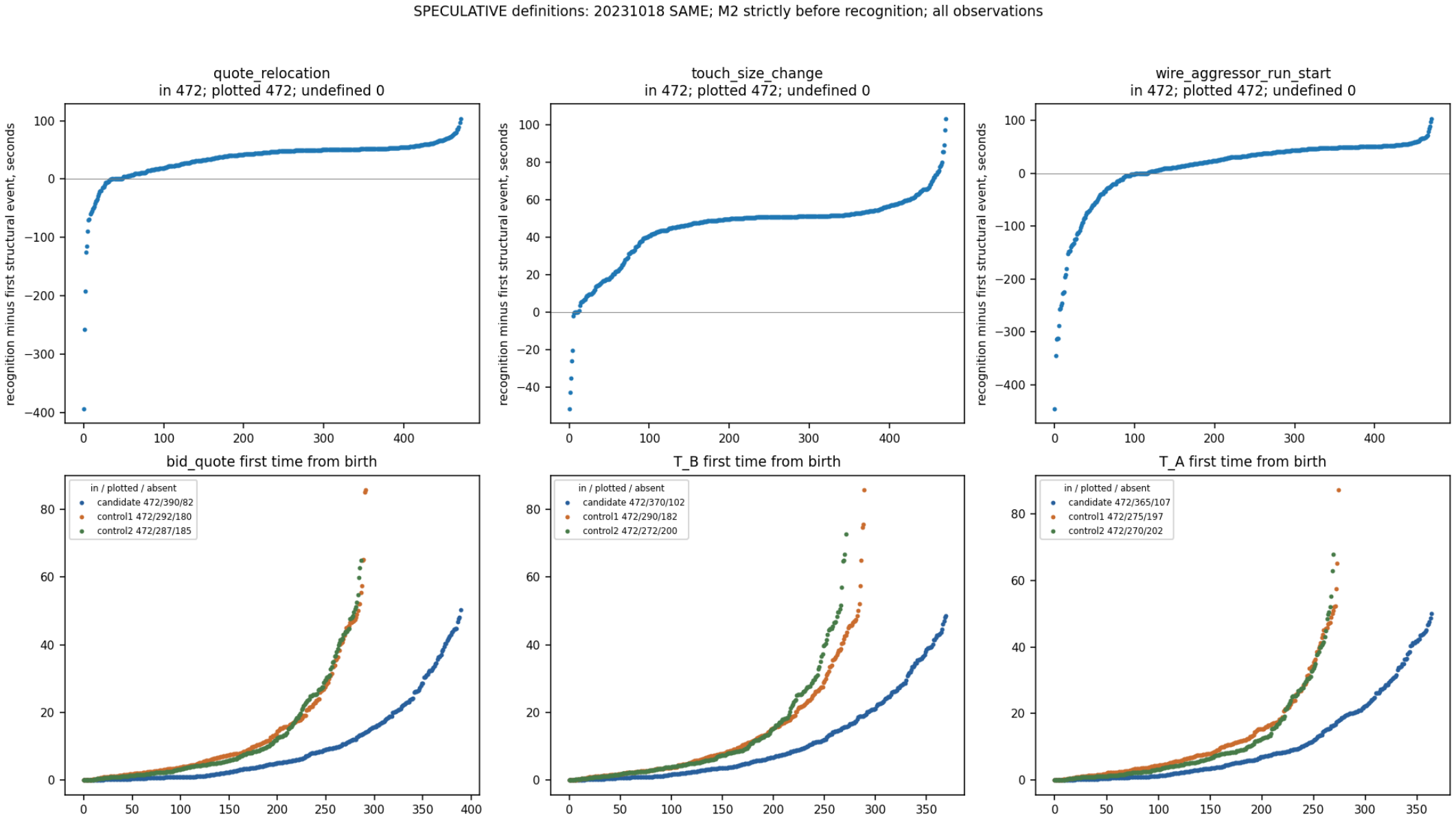

winning_offset_s
in 863; points 863; unavailable 0
ordered observation

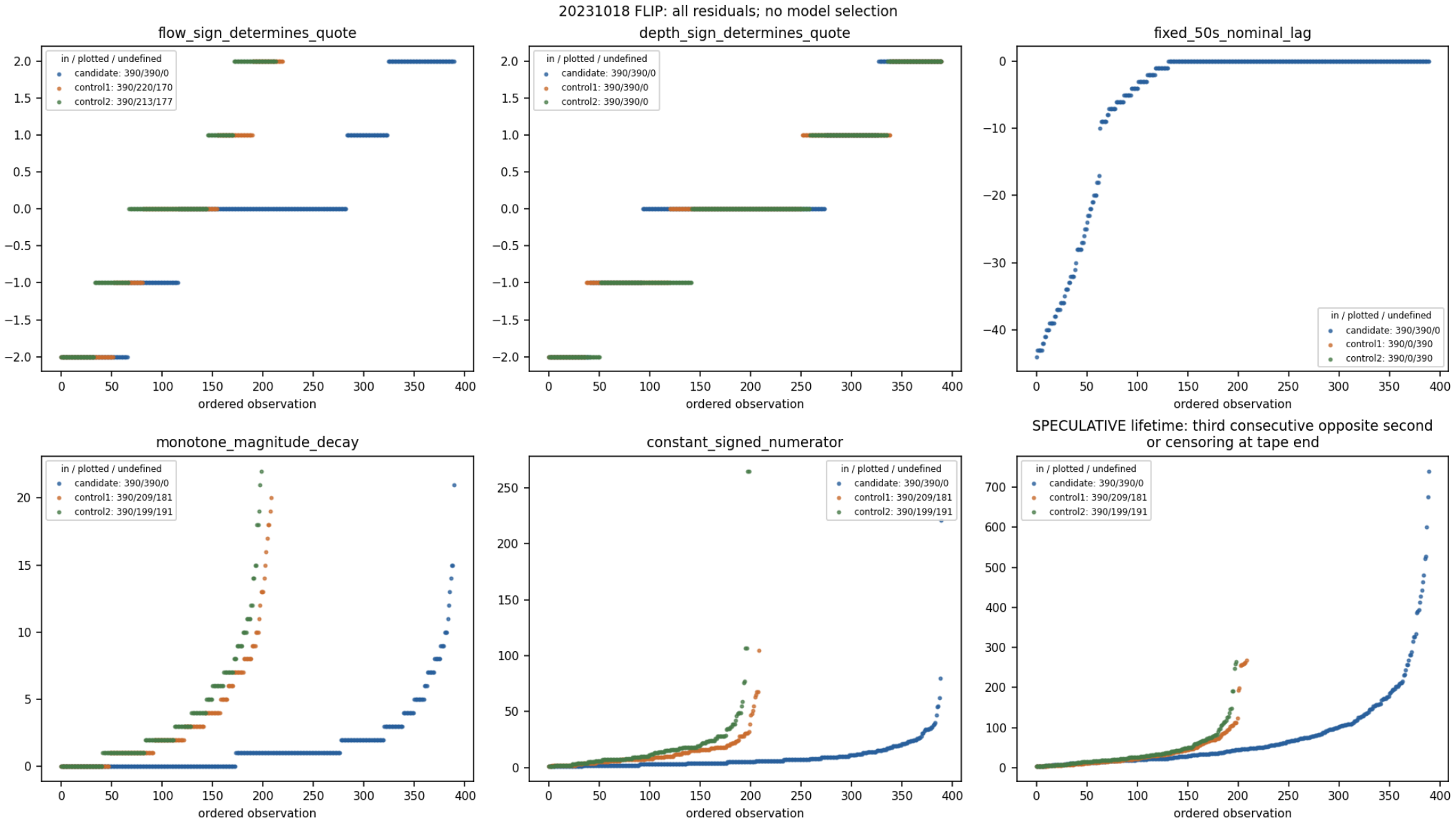

observations_behind_threshold
in 863; points 863; unavailable 0
ordered observation

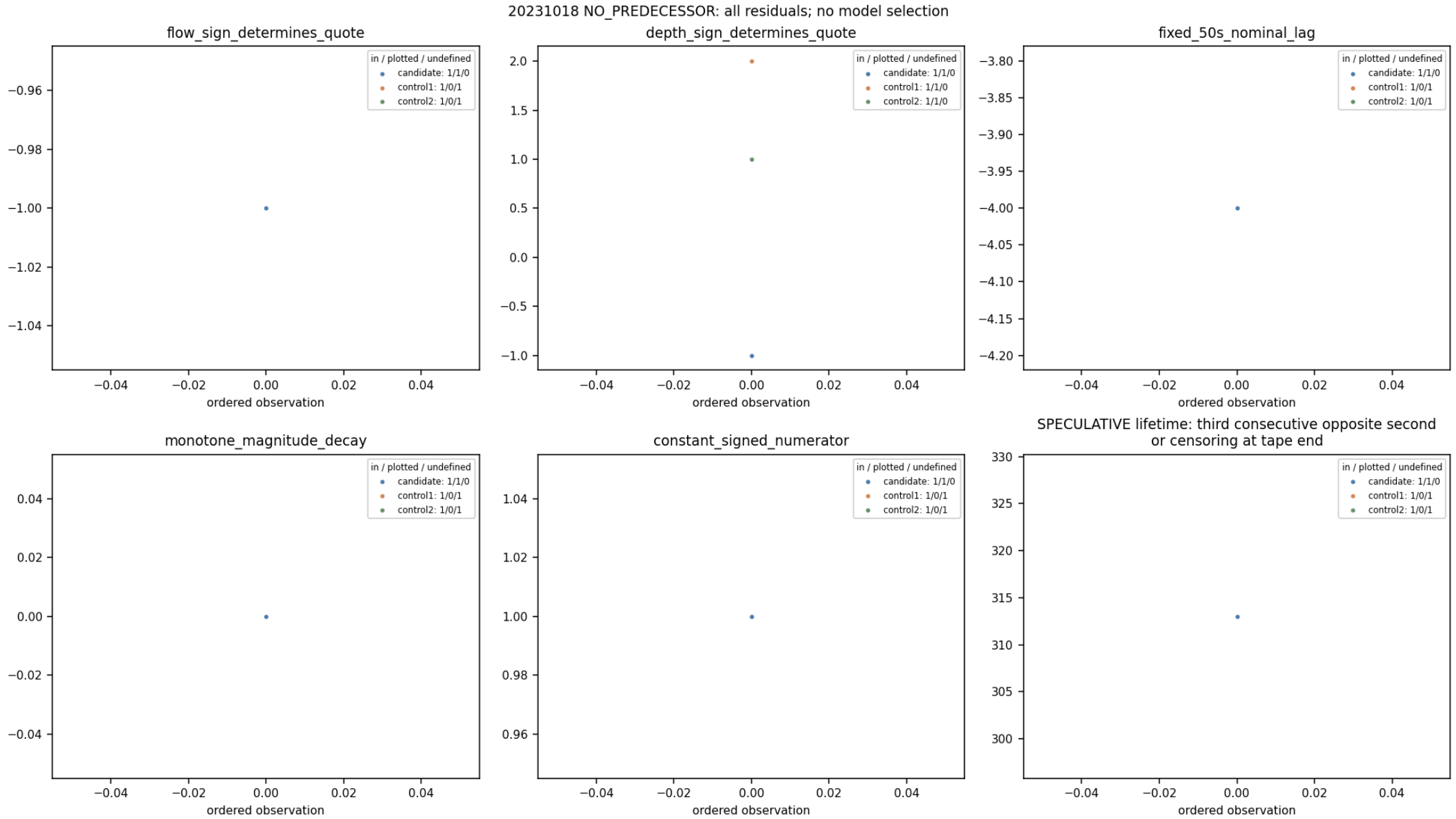

searched_span_seconds
in 863; points 863; unavailable 0
ordered observation


SPECULATIVE definitions: 20231018 FLIP; M2 strictly before recognition; all observations
quote_relocation
in 390; plotted 390; undefined 0
touch_size_change
in 390; plotted 390; undefined 0
wire_aggressor_run_start
in 390; plotted 390; undefined 0
recognition minus first structural event, seconds
bid_quote first time from birth
in / plotted / absent
candidate 390/323/67
control1 390/237/153
control2 390/237/153
T_B first time from birth
candidate 390/296/94
control1 390/230/160
control2 390/233/157
T_A first time from birth
candidate 390/299/91
control1 390/221/169
control2 390/236/154


SPECULATIVE definitions: 20231018 NO_PREDECESSOR; M2 strictly before recognition; all observations
quote_relocation
in 1; plotted 1; undefined 0
touch_size_change
in 1; plotted 1; undefined 0
wire_aggressor_run_start
in 1; plotted 1; undefined 0
recognition minus first structural event, seconds
bid_quote first time from birth
T_B first time from birth
T_A first time from birth
in / plotted / absent
candidate 1/0/1
control1 1/1/0
control2 1/0/1

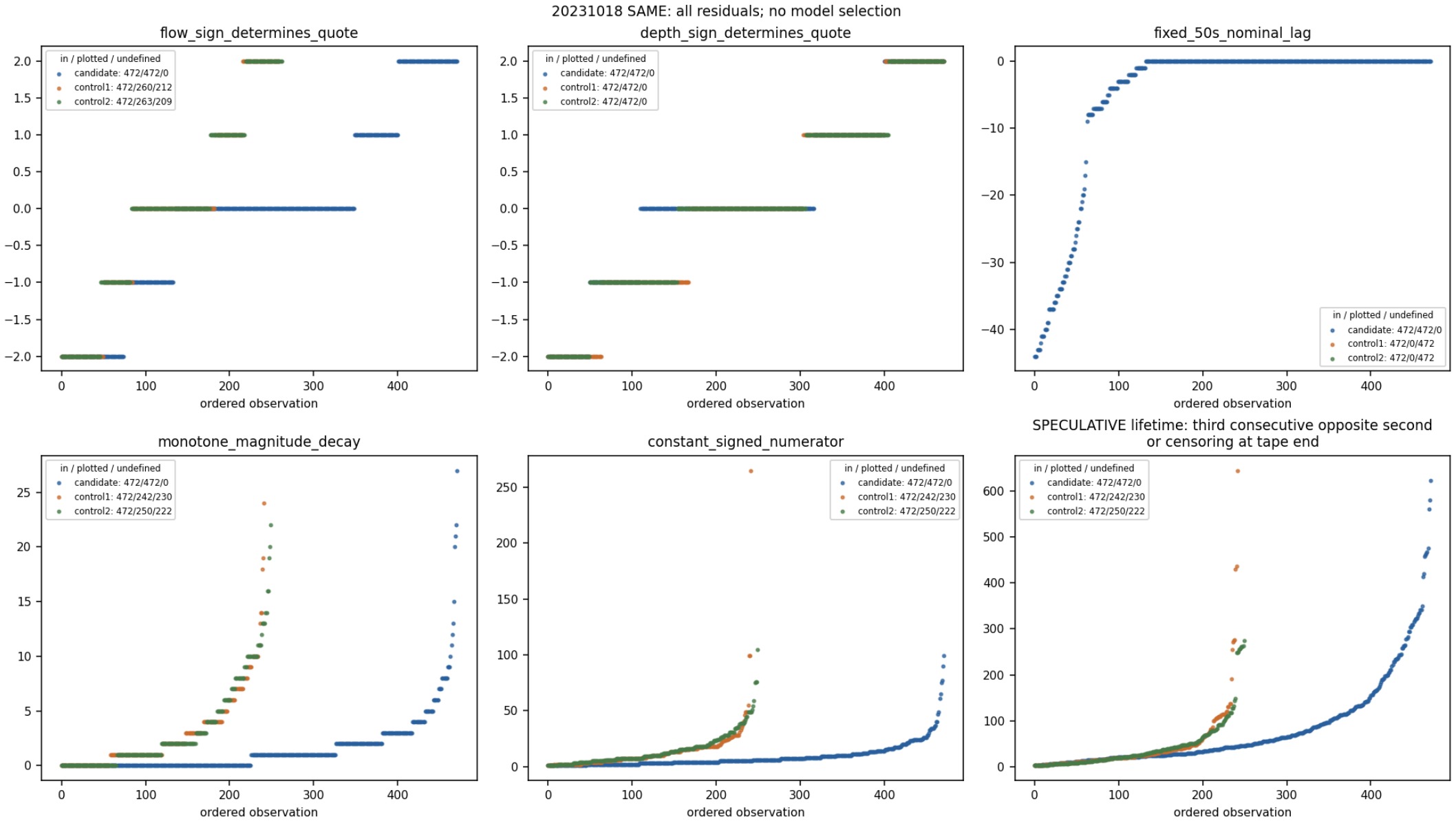

SPECULATIVE definitions: 20231018 SAME; M2 strictly before recognition; all observations
quote_relocation
in 472; plotted 472; undefined 0
touch_size_change
in 472; plotted 472; undefined 0
wire_aggressor_run_start
in 472; plotted 472; undefined 0
recognition minus first structural event, seconds
bid_quote first time from birth
T_B first time from birth
T_A first time from birth
in / plotted / absent
candidate 472/390/82
control1 472/292/180
control2 472/287/185
candidate 472/370/102
control1 472/290/182
control2 472/272/200
candidate 472/365/107
control1 472/275/197
control2 472/270/202


20231018 FLIP: all residuals; no model selection
flow_sign_determines_quote
depth_sign_determines_quote
fixed_50s_nominal_lag
monotone_magnitude_decay
constant_signed_numerator
SPECULATIVE lifetime: third consecutive opposite second or censoring at tape end
in / plotted / undefined
candidate: 390/390/0
control1: 390/220/170
control2: 390/213/177
control1: 390/390/0
control2: 390/390/0
control1: 390/0/390
control2: 390/0/390
control1: 390/209/181
control2: 390/199/191
ordered observation


20231018 NO_PREDECESSOR: all residuals; no model selection
flow_sign_determines_quote
depth_sign_determines_quote
fixed_50s_nominal_lag
monotone_magnitude_decay
constant_signed_numerator
SPECULATIVE lifetime: third consecutive opposite second or censoring at tape end
in / plotted / undefined
candidate: 1/1/0
control1: 1/0/1
control2: 1/0/1
control1: 1/1/0
control2: 1/1/0
ordered observation

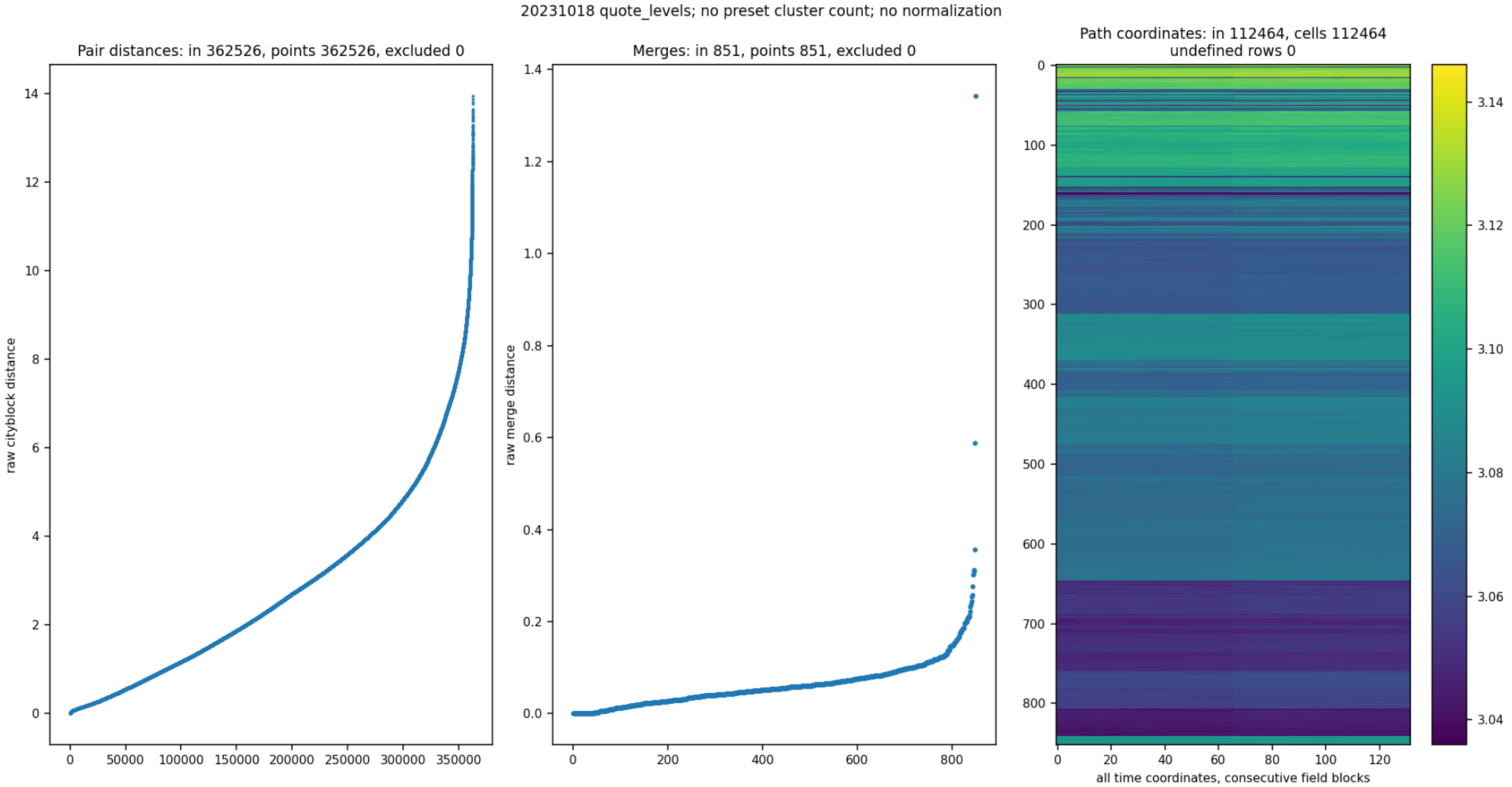
20231018-operators-SAME.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



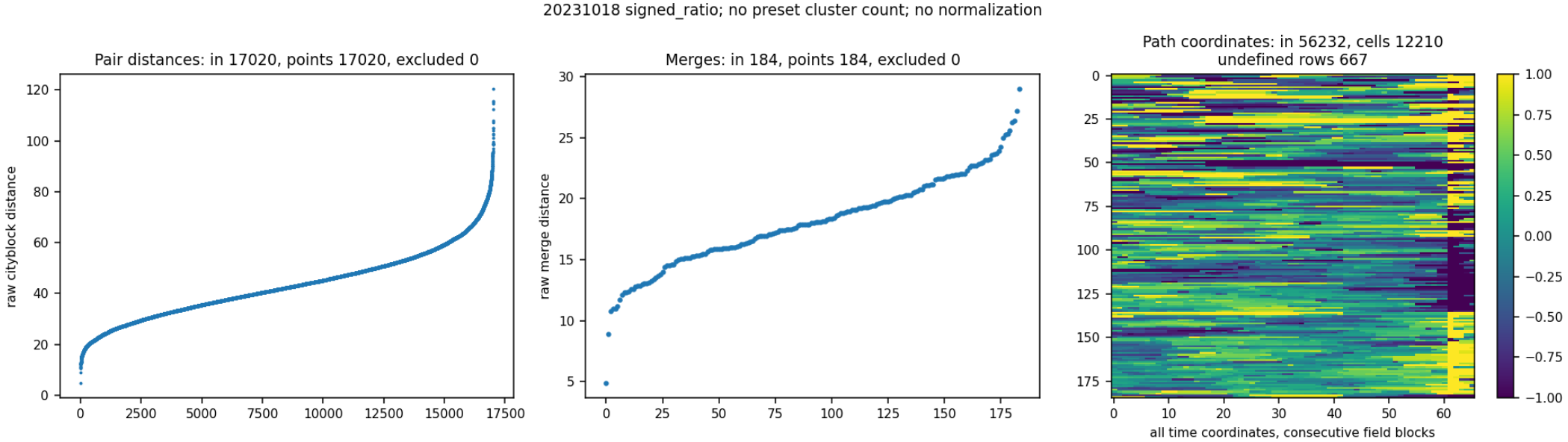
20231018-structure-quote_levels.png

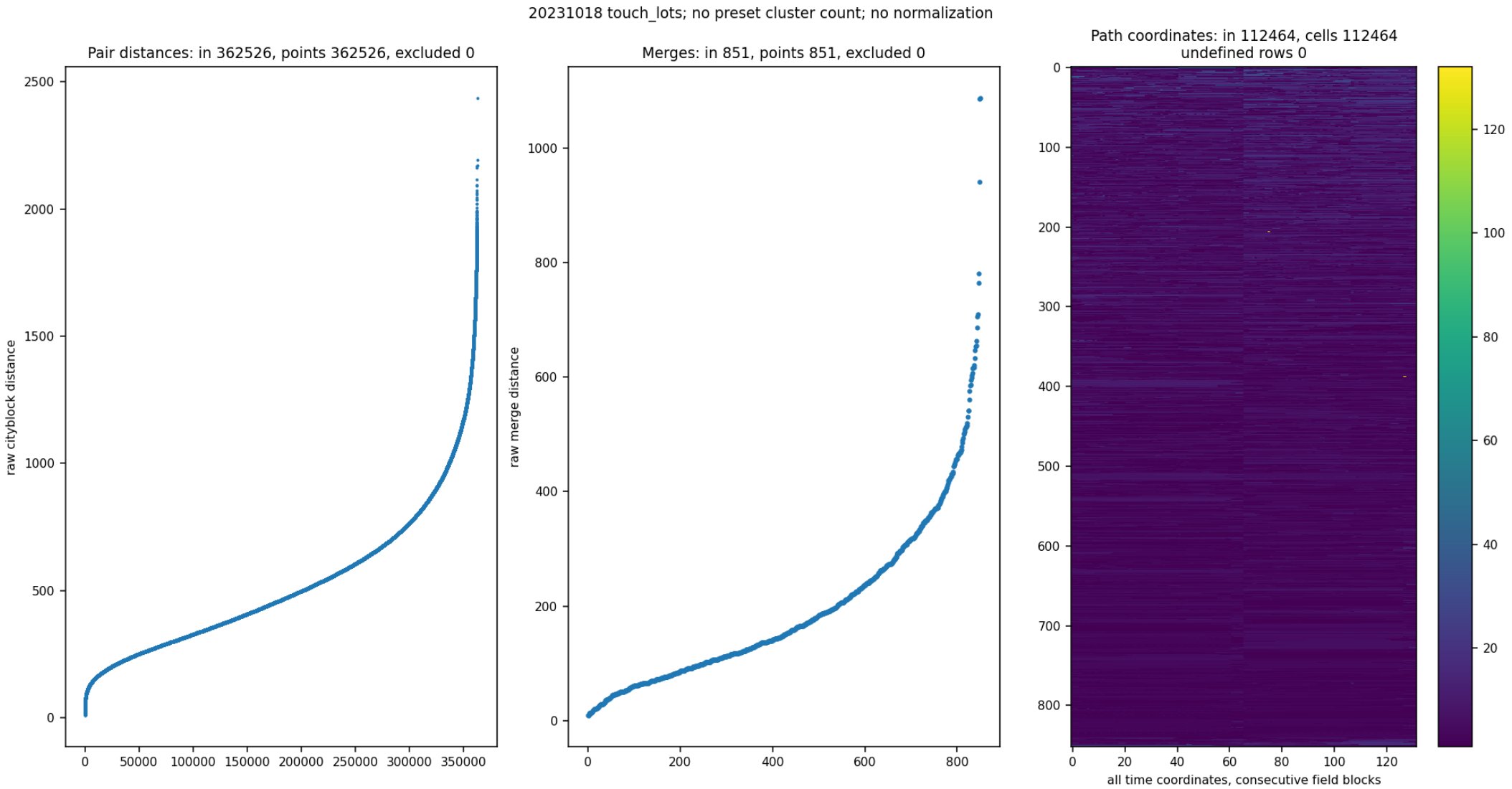
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

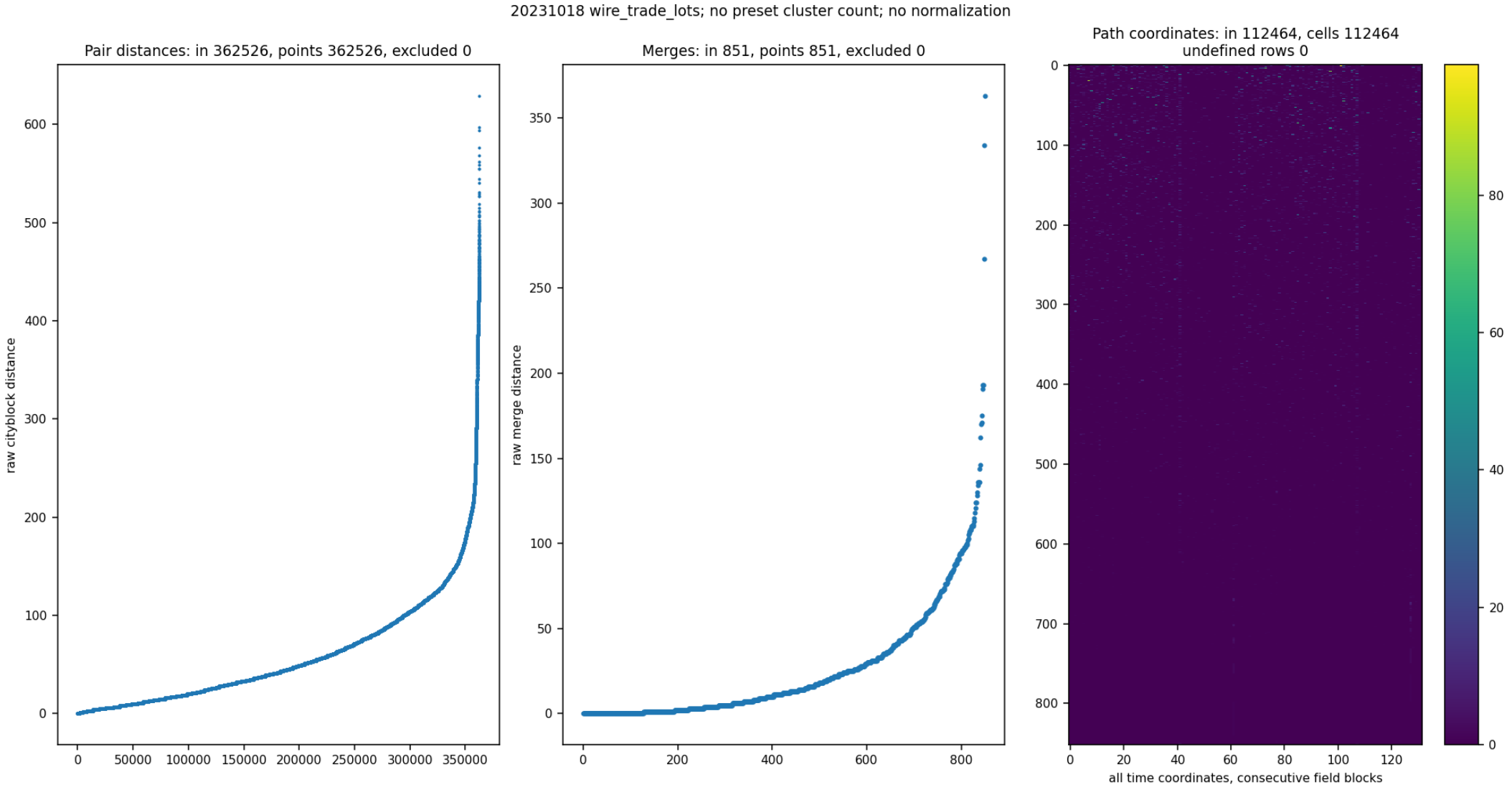


20231018-structure-signed_ratio.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.




20231018 touch_lots; no preset cluster count; no normalization
Pair distances: in 362526, points 362526, excluded 0
raw cityblock distance
Merges: in 851, points 851, excluded 0
raw merge distance
Path coordinates: in 112464, cells 112464
undefined rows 0
all time coordinates, consecutive field blocks


20231018 wire_trade_lots; no preset cluster count; no normalization
Pair distances: in 362526, points 362526, excluded 0
raw cityblock distance
Merges: in 851, points 851, excluded 0
raw merge distance
Path coordinates: in 112464, cells 112464
undefined rows 0
all time coordinates, consecutive field blocks